

A Project Report on

AN AUTOMATIC EARTHQUAKE FORECASTING

SYSTEM USING MACHINE LEARNING APPROACH

A Thesis submitted in partial fulfilment of the requirement for the award of
the degree of

Bachelor of Technology in Computer Science and Engineering

by

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

JNTU-GV COLLEGE OF ENGINEERING, VIZIANAGARAM

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(2019-2023)



Department of Computer Science & Engineering

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CERTIFICATE

This is to certify that the dissertation entitled “**AN AUTOMATIC EARTHQUAKE FORECASTING SYSTEM USING MACHINE LEARNING APPROACH**” that is being submitted by **CH.BHAVYA MEGHANA (19VV1A0509), B.NEELIMA CHOWDARY (20VV5A0567), V.KOMALA (20VV5A0568), M.HARSHA VARDHAN (19VV1A0540)**, in partial fulfilment for the award of Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING to the JNTU-GV College of Engineering, Vizianagaram is a record of Bonafide work carried them under our Guidance and Supervision.

The results embodied in this dissertation have not been submitted to any other University or Institute for the award of any degree or diploma.

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DECLARATION

We, **CH.BHAVYA MEGHANA, B.NEELIMA CHOWDARY, V.KOMALA, M.HARSHA VARDHAN** hereby declare that the project report titled "**AN AUTOMATIC EARTHQUAKE FORECASTING SYSTEM USING MACHINE LEARNING APPROACH** " submitted to JNTU-GV College Of Engineering, Vizianagaram, in partial fulfilment of the requirements for the award of the degree of B.Tech. in COMPUTER SCIENCE AND ENGINEERING is award of original and independent research work done by us during the academic year 2022 - 2023 under the supervision of **Dr. R. RAJESWARA RAO** and it has not formed the basis of any Degree/Diploma/Associateship/ Fellowship or other similar title to any candidate in University.

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PSO3: Able to be a technically competent employee, researcher, entrepreneur, excel in competitive exams and zest for higher studies.

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It is needed with a great sense of pleasure and immense sense of gratitude that we acknowledge the help of these individuals. We owe many thanks to many people who helped and supported us during the writing of this report.

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ABSTRACT

An earthquake is caused by a sudden slip on a fault .The tectonic plates are always slowly moving , but they get stuck at their edges due to friction. When the stress on the edge overcomes the friction, there is an earthquake that releases energy in waves that travel through the earth's crust and cause the shaking that we feel .The primary effects of earthquakes are ground shaking, ground rupture, landslides ,tsunamis and liquefaction. Many buildings, hospitals, schools, etc are destroyed because of it. Lot of people got injured and lost their properties.

So here we have currently focused on scientific study related to earthquake prediction or forecasting which focuses on when and where it will occur. As the earthquakes are spatially and temporally correlated because of the crust movement, we can forecast the earthquakes based on that real-time data.

The goal of Earthquake Forecasting is to give warning of potentially damaging earthquakes early enough to allow appropriate response to the disaster, enabling people to minimize loss of life and property.

The project entitled “An Automatic Earthquake Forecasting System Using Machine Learning Approach” is mapped with following PO’S and PSO’S:

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12

PSO1	PSO2	PSO3

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CHAPTER-1

INTRODUCTION

1. INTRODUCTION

Countless dollars and entire scientific careers have been dedicated to predicting where and when the next big earthquake will strike. But unlike weather forecasting, which has significantly improved with the use of better satellites and more powerful mathematical models, earthquake prediction has been marred by repeated failure due to highly uncertain conditions of earth and its surroundings.

Now, with the help of artificial intelligence, a growing number of scientists say changes in the way they can analyze massive amounts of seismic data can help them better understand earthquakes, anticipate how they will behave, and provide quicker and more accurate early warnings.

This helps in hazard assessments for many builders and real estate business for infrastructure planning from business perspective. Also, many lives can be saved through early warning. This project aims a simple solution to above problem by predicting or forecasting likely places to have earthquake in next 7 days. For user-friendly part, this project has a web application that extracts live data updated every minute by USGS.gov and predicts next likely place world-wide to get hit by an earthquake, hence a real-time solution is provided.

1.1 WHAT IS AN EARTHQUAKE ?

An earthquake is a trembling or a shaking movement of the ground, caused by the slippage or rupture of a fault within the Earth's crust. A sudden slippage or rupture along a fault line results in an abrupt release of elastic energy stored in rocks that are subjected to great strain.

This energy can be built up and stored over a long time and then released in seconds or minutes. Strain on the rocks results in more elastic energy being stored which leads to far greater possibility of an earthquake event. The sudden release of energy during an earthquake causes low-frequency sound waves called seismic waves to propagate through the Earth's crust or along its surface.

Every year more than 3 million earthquakes take place, most of these unnoticed by humans. In contrast, a severe earthquake is the most frightening and catastrophic event of nature which can occur anywhere on the surface of our planet!

Although usually lasting only seconds, a severe earthquake in a densely populated area may have catastrophic effects causing the death of hundreds of thousands of people, injuries, destruction and enormous damage to the economies of the affected area.



Fig:1.1: Damage caused by Earthquake

Hundreds of thousands of people have been killed by earthquakes despite scientists being able to predict and forewarn in advance and engineers construct earthquake-safe buildings. Unfortunately earthquakes occur often in countries which are unable to afford earthquake-safe construction.

Besides the immediate, obvious threat presented by an earthquake, it can also set off several other natural hazards. The energy release resulting from earthquakes can easily trigger slope failures. A tsunami may be formed which causes flood on coastal areas. These events occur along with volcanic activity, resulting in even more potential danger.

1.2 WHAT CAUSES EARTHQUAKES?

The tectonic plates that make up the earth's crust are moving constantly. As the edges of these plates slide against each other in fault zones, friction can slow them down, leading to the buildup of pressure over long periods of time. When the force of movement finally overcomes the friction, sections of the crust suddenly break or become displaced, releasing the pent-up pressure in the form of seismic waves. This is a naturally occurring earthquake, sometimes referred to as a **tectonic earthquake**.

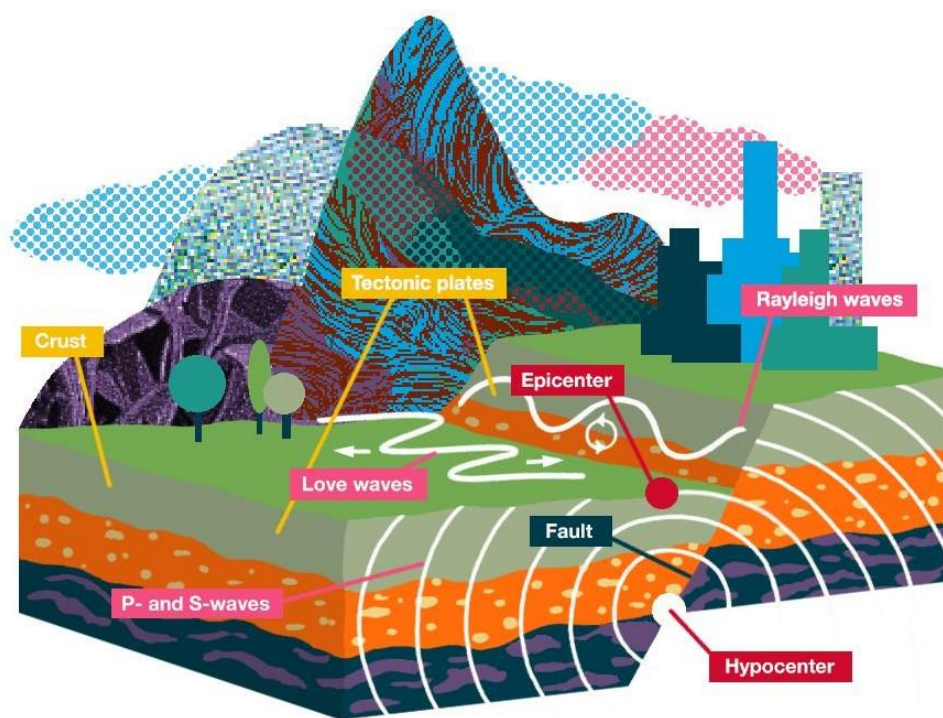


Fig 1.2: Earthquake parameters

While tectonic earthquakes can occur at any location around the world, the majority of large earthquakes—about 80 percent—occur at the circum-Pacific seismic belt found along the rim of the Pacific Ocean. Two other regions regularly exhibiting earthquakes include the Alpide belt, extending along the southern margin of Eurasia through the Himalayan Mountains, Sumatra, and Java; and the Mid-Atlantic Ridge running along the floor of the Atlantic Ocean.

Also there are other causes like:

1. Groundwater Extraction.
2. Any disturbance on the surface.
3. Big Skyscrapers.
4. Tracking and injecting wells.

Earthquakes is one of the worst natural hazards which often turn into disaster causing widespread destruction and loss to human life.

Recently an Earthquake of Magnitude:3.4 occurred on 25-08-2022 of Kolhapur, Maharashtra, India.

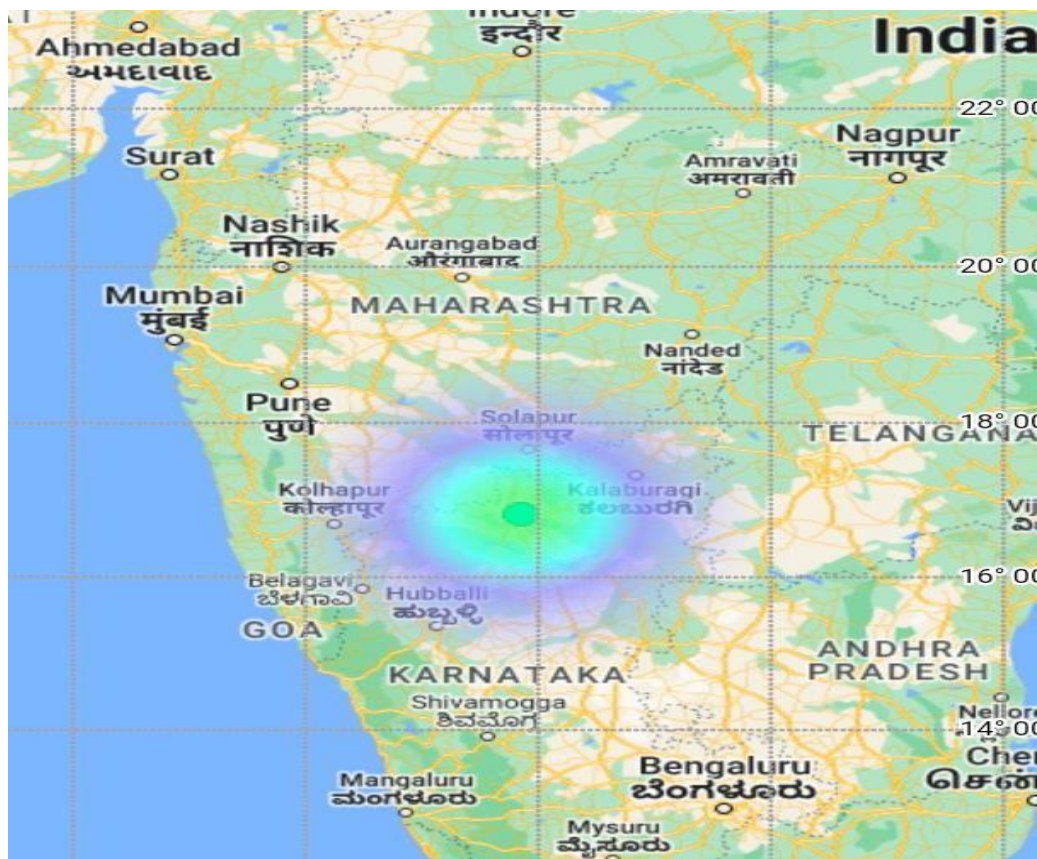


Fig 1.3 : Earthquake held at Maharastra

The National Earthquake Information Centre (NEIC) records an average of 20,000 earthquakes every year (about 50 a day) around the world.

So, it became the focus of interest for a variety of fields, like seismological research and environmental engineering.

1.3 Worldwide Earthquakes from 2000 to 2021

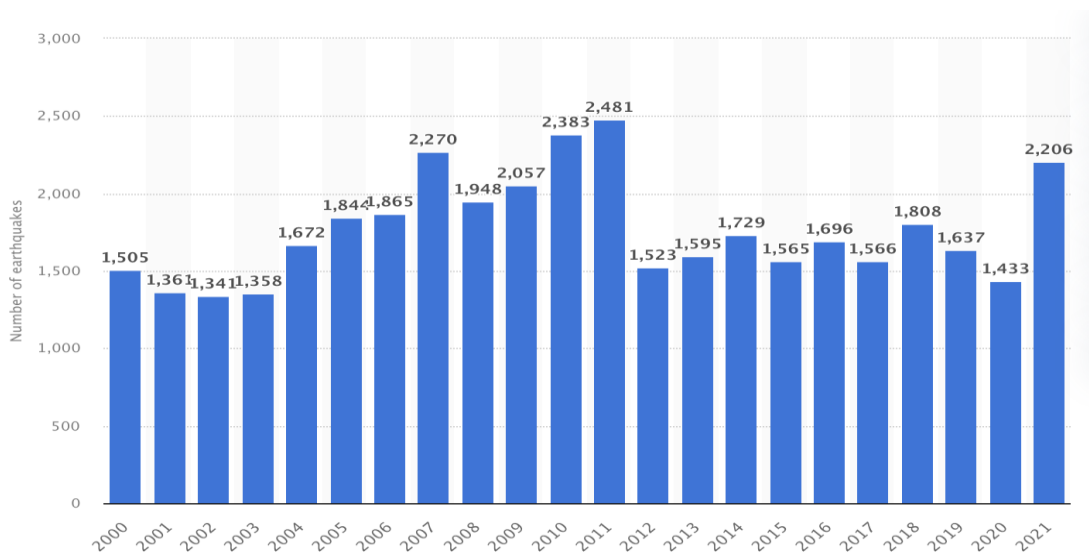


Fig 1.4 :Worldwide Earthquake from 2000 to 2021

1.4 Machine Learning

Machine learning is programming computers to optimize a performance criterion using example data or past experience. We have a model defined up to some parameters, and learning is the execution of a computer program to optimize the parameters of the model using the training data or past experience.

The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data.

The field of study known as machine learning is concerned with the question of how to construct computer programs that automatically improve with experience.

Machine learning is a subfield of artificial intelligence that involves the development of algorithms and statistical models that enable computers to improve their performance in tasks through experience.

These algorithms and models are designed to learn from data and make predictions or decisions without explicit instructions. There are several types of machine learning, including supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabelled data. Reinforcement learning involves training a model through trial and error.

Machine learning is used in a wide variety of applications, including image and speech recognition, natural language processing, and recommender systems.

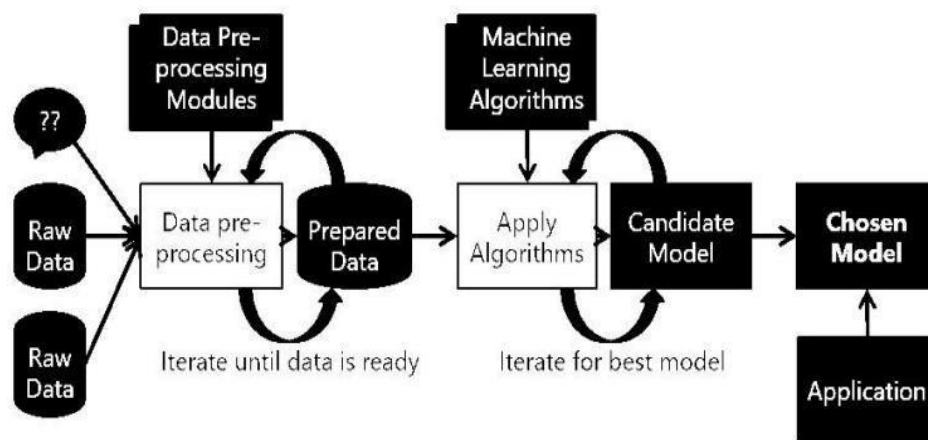


Fig 1.5 : Methodology Of Machine Learning Model

CHAPTER 2

LITERATURE SURVEY

2. LITERATURE SURVEY

The purpose of a literature review is to gain an understanding of the existing research and debates relevant to a particular topic or area of study, and to present that knowledge in the form of a written report. Conducting a literature review helps you build your knowledge in your field. You'll learn about important concepts, research methods, and experimental techniques that are used in your field. You'll also gain insight into how researchers apply the concepts you're learning in your unit to real world problems. Another great benefit of literature reviews is that as you read, you'll get a better understanding of how research findings are presented and discussed in your particular discipline. If you pay attention to what you read and try to achieve a similar style, you'll become more successful at writing for your discipline.

1. EXISTING MODELS

1.1. Major earthquake event prediction using various machine learning algorithms [1]

Authors: Roxane Mallouhy, Chady Abou Joude, Christophe Guyeux, Abdhallah Makhoul.

Here in this paper, KNN, Random Forest and MLP are the best by producing the least false output (FP) while SVM, KNN, MLP and Random Forest classify the higher number of output correctly. The study has been applied to a dataset collected from a center in California, which was recording inputs for 36 years . Generally here there exist at least two basic categories of earthquake prediction : Short-term predictions and forecast ones. Short term earthquake predictions are made hours or days in advance, while forecasts are predicted months to years in advance. The majority of studies are done on forecast, taking into consideration the history of earthquakes in specific countries and areas. In this context, the core idea of this work is to predict whereas an event is classified as negative or positive major earthquake by applying different machine learning algorithms.

Eight different algorithms have been applied on a real earthquake dataset, namely: Random Forest, Naive Bayes, Logistic Regression, MultiLayer Perceptron, AdaBoost, K-nearest neighbors, Support Vector Machine, and Classification and Regression Trees. For each selected model, various hyperparameters have been selected, and obtained prediction results have been fairly compared using various metrics, leading to a reliable prediction of major events for 3 of them.

1.2. Time Series Analysis on Earthquakes Using EDA and Machine Learning[2]

Authors: Muhammad Fakhrillah Abdul Azis, Fariz Darari; Muhammad Rizqy Septyandy.

In this paper, they proposed an approach to time series analysis over earthquake data, consisting of two steps: exploration and prediction. They had first performed exploration step which relies on exploratory data analysis (EDA) comprising descriptive statistics and data visualization, after that they performed prediction step that focused on how to predict the number of earthquakes for the following years. They performed our time series analysis using various machine learning techniques over a global earthquake dataset from 1965-2016 and report insights as well as lessons learned from the study.

1.3. Analysis of earthquake forecasting using random forest [3]

Authors: Kholiq Budiman, Yahya Nur Ifriza

In this work, they have attempted for forecasting earthquakes before they occur using p-arrival data, which includes information on disaster arrival time and amplitude height from the arrival station. They have performed several studies on earthquake prediction that have been carried out so far and have developed and used the Random Forest method and one of the Machine Learning. They have conducted a random forest prediction model to identify the structural safety status of buildings damaged by the earthquake is probabilistic. And An earthquake's latitude, longitude, magnitude, and depth are predicted using the random forest algorithm.

This study's predictions were accurate to within 63 percent achieved an F-score of 80% and 79% for binary classification and 64% and 60% for ternary classification, respectively.

2. MOTIVATION

There are several motivations for forecasting earthquakes, even if it is uncertain whether they will occur or not. Here are a few reasons:

Public Safety: By providing advanced warning or preparing for potential earthquake events, authorities can take measures to minimize the damage and loss of life caused by an earthquake.

Infrastructure Planning: Forecasting earthquakes can help engineers and planners design and construct buildings, bridges, and other infrastructure that can withstand seismic activity.

Scientific Research: Studying earthquakes and their patterns can provide valuable insights into the Earth's interior and the processes that occur there. It can also help scientists better understand the nature of earthquakes and potentially lead to new technologies and methods for predicting them.

Emergency Preparedness: Even if an earthquake does not occur, forecasting can be used to prepare emergency responders, hospitals, and other organizations for a potential event.

As earthquake prediction models are not able to predict early and accurately .It might lead to disastrous effects on humans and environment such as property damage , loss of life etc.

To overcome this loss, we are proposing An Automatic Earthquake Forecasting system that is used for forecasting earthquake possible location.

CHAPTER 3

MACHINE LEARNING BASED MODEL FOR FORECASTING AN EARTHQUAKE

3. MACHINE LEARNING BASED MODEL FOR FORECASTING AN EARTHQUAKE

In this chapter we are going to discuss the steps that are to be followed in order to build an automatic earthquake forecasting model. The methodology gives the detailed description of our model that supports our findings by explaining our proposed techniques and creating a roadmap of how we have reached our conclusions. In the section 3.1 the proposed methodology is discussed. How the raw data is preprocessed and cleaned by using preprocessing techniques is discussed in section. In the section 3.2 we have explained how the data used was collected. In the section 3.3 we have discussed the preprocessing of data according to the considered project. In the section 3.4 we have discussed how the feature engineering have taken place and what are the uses and needs of feature engineering is explained and then in section 3.5 we have discussed the training phases that is about the models and their algorithms and why only those algorithms are all explained and also the mathematical analysis of each algorithms are explained.

1.METHODOLOGY USED

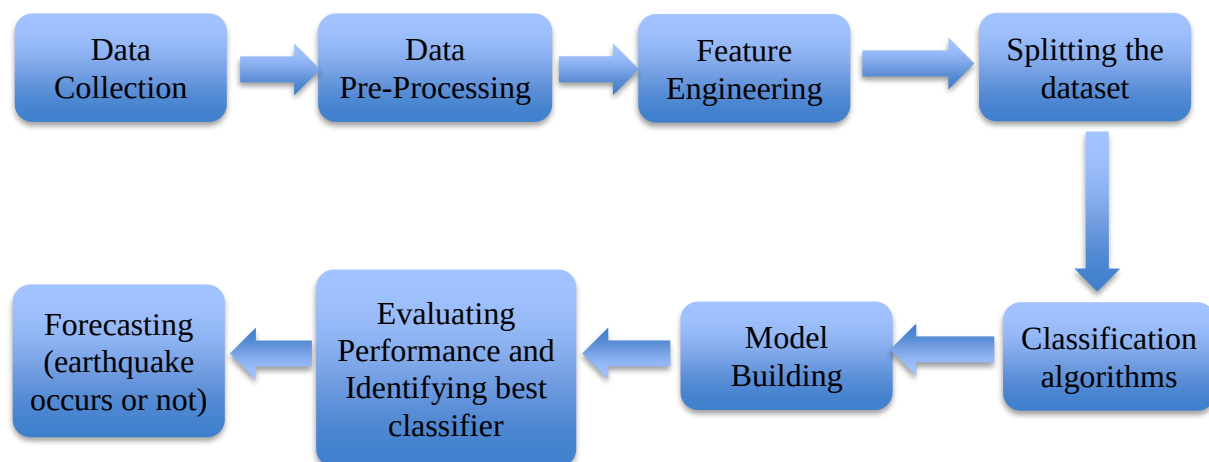


Fig. 3.1: System Architecture

2.DATA COLLECTION

We have collected the earthquake dataset from USGS(United States Geological Survey).It is a real time data that updates every minute.

- Collecting data allows you to capture a record of past events so that we can use data analysis to find recurring patterns.
- Below is the link for the considered dataset.

<https://earthquake.usgs.gov/earthquakes/feed/v1.0/csv.php>

all_month (1)																					
time	latitude	longitude	depth	mag	magType	net	gap	dmin	rms	net_id	updated	place	type	horizontalError	depthError	magError	maght	status	locationSource	magSource	
2023-02-08T09:57:27.020Z	38.83151	-122.816197	1.78	1.06	mb	21	51	0.006891	0.01	nc73843476	2023-02-08T09:59:04.016Z	8km NW of The Geysers, CA	earthquake	0.18	0.44	0.08	20	automatic	nc		
2023-02-08T09:53:59.430Z	36.9614983	-121.6100008	7.34	1.24	mb	8	72	0.06158	0.06	nc73843456	2023-02-08T09:46:10.865Z	6km SW of Gilroy, CA	earthquake	0.47	1.06	0.33	10	automatic	nc		
2023-02-08T09:29:23.110Z	62.8838	-150.5066	56.2	1.7	ml				0.26	ak	ak0231anbpt1	2023-02-08T09:31:24.033Z	45 km NNE of Petersville, Alaska	earthquake			0.6		automatic	ak	
2023-02-08T09:27:15.110Z	36.9613342	-121.598899	5.03	1.24	mb	11	76	0.05883	0.06	nc73843451	2023-02-08T09:55:12.601Z	6km SSW of Gilroy, CA	earthquake	0.38	0.93	0.37	10	automatic	nc		
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2023-02-08T09:19:32.372Z	59.9782	-153.2467	120.5	2.4	ml				0.56	ak	ak0231anbth1	2023-02-08T09:19:13.322Z	52 km ENE of Pedro Bay, Alaska	earthquake		0.5			automatic	ak	
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2023-02-08T09:06:31.350Z	33.5775	-116.718	2.19	0.42	ml	10	138	0.02149	0.15	ci	ci04912176	2023-02-08T09:10:05.190Z	5km WNW of Anza, CA	earthquake	0.74	0.37	0.195	9	automatic	ci	
2023-02-08T09:06:27.880Z	38.8409996	-122.876003	2.26	0.78	mb	13	120	0.00695	0.02	nc73843431	2023-02-08T09:09:17.284Z	13km NW of The Geysers, CA	earthquake	0.55	0.63	0.19	17	automatic	nc		
2023-02-08T07:51:09.607Z	59.9633	-152.6905	61.3	2	ml				1.05	ak	ak0231anbfg1	2023-02-08T07:53:24.526Z	54 km SW of Narsarsuk, Alaska	earthquake		1.4			automatic	ak	
2023-02-08T07:50:14.630Z	19.1353342148896	-155.37483215332	29.5330000695455	2.11900069	mb	40	184			h77321212	2023-02-08T07:53:24.560Z	13 km SE of Pahala, Hawaii	earthquake	0.66	1.25	0.730000019	17	automatic	hv		
2023-02-08T07:48:32.134Z	37.9993	36.4734	10	4.9	mb	70	24	0.983	0.96	us	us0000mg1	2023-02-08T07:50:27.075Z	13 km SW of Gökçus, Turkey	earthquake	4.11	1.857	0.082	47	reviewed	us	us
2023-02-08T07:40:30.510Z	38.7904985	-122.7849994	0.86	0.75	mb	10	102	0.006781	0.01	nc73843426	2023-02-08T07:37:29.105Z	1km WNW of The Geysers, CA	earthquake	0.29	0.46	0.18	9	automatic	nc		
2023-02-08T07:45:00.736Z	59.9576	-152.2637	61.4	1.9	ml				0.62	ak	ak0231anbth1	2023-02-08T07:47:48.458Z	28 km WNW of Anchor Point, Alaska	earthquake		0.7			automatic	ak	
2023-02-08T07:36:10.307Z	62.9274	-161.2687	114.7	1.6	ml				0.12	ak	ak0231anbcb1	2023-02-08T07:36:36.462Z	55 km NNW of Petersville, Alaska	earthquake		1.3			automatic	ak	
2023-02-08T07:34:35.816Z	62.945	-149.8879	58.2	1.6	ml				0.22	ak	ak0231anbcb1	2023-02-08T07:36:13.992Z	15 km SSE of Sleetina North, Alaska	earthquake		0.9			automatic	ak	
2023-02-08T07:33:23.130Z	38.7849884	-122.738325	0.59	2.16	ml	37	89	0.01302	0.04	nc73843421	2023-02-08T09:46:10.547Z	3km NE of The Geysers, CA	earthquake	0.16	0.37	0.19	38	automatic	nc		
2023-02-08T07:21:47.630Z	34.9055	-116.5451967	2.62	1.26	ml	14	175	0.02027	0.3	ci	ci04912166	2023-02-08T07:25:29.694Z	14km WNW of Maltus, CA	earthquake	1.11	0.65	0.211	21	automatic	ci	
2023-02-08T07:11:15.170Z	18.181	-47.989	93.77	3.43	mb	11	324	0.033	0.18	pr	pr71355608	2023-02-08T07:44:12.220Z	63 km SE of Punta Cana, Dominican Republic	earthquake	1.95	1.08	0.28328668955113	6	reviewed	pr	pr
2023-02-08T06:53:50.630Z	19.2099999755654	-155.413497924802	32.2799997702999	2.63	ml	45	120			h773212177	2023-02-08T06:28:03.040Z	9 km NE of Pahala, Hawaii	earthquake	0.49	0.589999974	3.11	28	automatic	hv		
2023-02-08T06:30:08.360Z	38.7999992	-122.8188324	10.91	0.77	mb	6	213	0.01199	0.02	nc73843411	2023-02-08T07:41:32.700Z	6km WNW of The Geysers, CA	earthquake	3.03	7.56	0.4	8	automatic	nc		
2023-02-08T06:30:01.847Z	42.9888	145.3592	53.335	4.4	mb	61	134	1.887	0.41	us	us0000mfs	2023-02-08T06:02:42.040Z	41 km SSW of Niiumun, Japan	earthquake	8.87	7.583	0.072	56	reviewed	us	us
2023-02-08T06:27:07.090Z	38.7911682	-122.7210007	3.44	0.78	mb	7	96	0.01462	0.01	nc73843406	2023-02-08T07:45:45.627Z	3km NW of Anderson Springs, CA	earthquake	0.45	2.83	0.05	8	automatic	nc		
2023-02-08T06:26:36.255Z	5.8244	-82.6747	10	4.6	mb	34	224	2.841	0.43	us	us0000mfr	2023-02-08T07:41:52.040Z	287 km S of Barica, Panama	earthquake	8.83	1.948	0.133	19	reviewed	us	us
2023-02-08T06:19:47.871Z	61.3375	-141.943	4.8	1.5	ml				0.42	ak	ak0231anbva1	2023-02-08T06:23:01.540Z	53 km ESE of McCarthy, Alaska	earthquake		0.6			automatic	ak	
2023-02-08T06:15:49.077Z	62.6073	-161.0868	79.1	2.1	ml				0.49	ak	ak0231anbuz1	2023-02-08T06:18:15.944Z	23 km NW of Petersville, Alaska	earthquake		0.5			automatic	ak	
2023-02-08T06:13:11.840Z	38.7903547	-122.7289963	-0.92	0.99	mb	6	130	0.01218	0.07	nc73843401	2023-02-08T06:30:11.271Z	3km NE of The Geysers, CA	earthquake	0.35	2.87	0.22	7	automatic	nc		
2023-02-08T06:11:04.362Z	61.1348	-161.9452	95.6	2.3	ml				0.40	ak	ak0231anb3n	2023-02-08T06:13:28.040Z	44 km W of Yoncos, Alaska	earthquake		0.9			automatic	ak	
2023-02-08T06:10:05.440Z	38.7949982	-122.7289977	1.03	1.13	mb	9	103	0.01297	0.02	nc73843399	2023-02-08T06:30:10.474Z	3km NE of The Geysers, CA	earthquake	0.26	0.79	0.26	9	automatic	nc		
2023-02-08T06:09:01.807Z	38.3573	36.1303	10	4.4	mb	39	186	1.158	0.59	us	us0000mfr	2023-02-08T07:09:30.941Z	Turkey-Syria border region	earthquake	4.45	1.965	0.11	24	reviewed	us	us

Fig 3.2 : Dataset

3.DATA PRE-PROCESSING

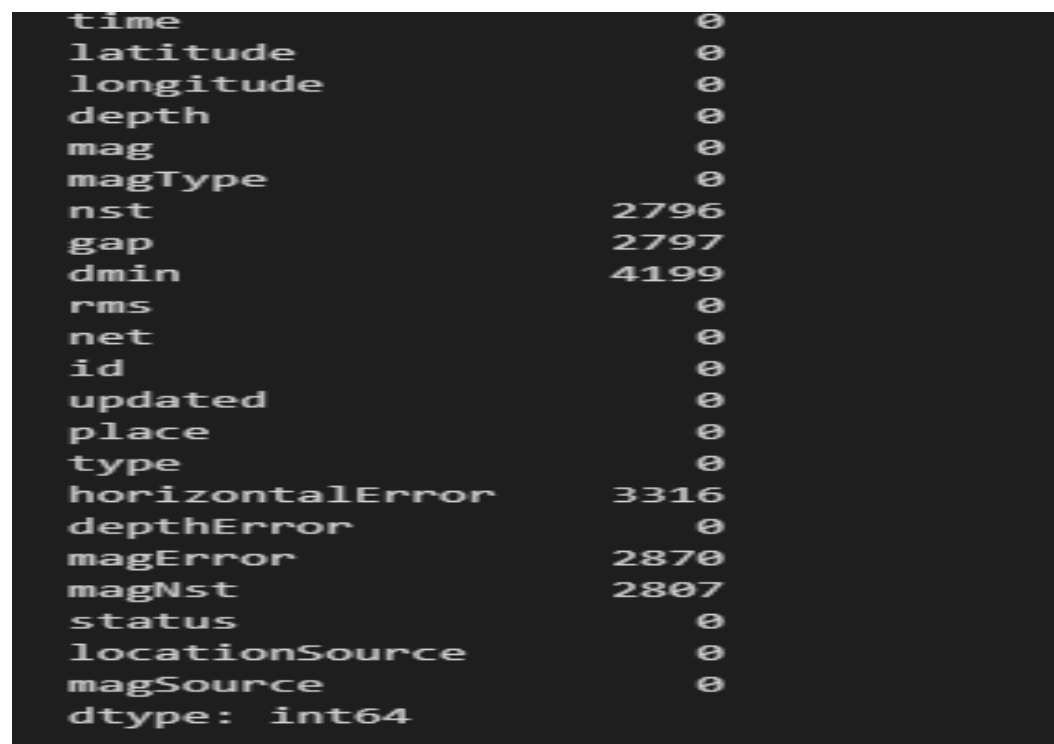
Data preprocessing is a way of converting the raw data into a much-desired form so that useful information can be derived from it, which is fed into the training model for successful decisions. Data pre-processing is a data mining technique that involves transforming raw data into an understandable format. Real-world data is often incomplete, inconsistent, and/or lacking in certain behaviour or trends, and likely to contain many errors. Data pre-processing is used data-driven data base application such as customer relationship management and rule-based applications.

3.1 WHAT IS THE NEED OF DATA PREPROCESSING

A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for machine learning models. Data pre-processing is required for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning model.

3.2 Null values: Input to model from dataset has many important features to consider as time, latitude & longitude, depth of quake, magnitude, place, rest other features are error and non supporting features for classification.

As there are lots of null values of certain features, but as part of prediction most of the features that address 'error' in measurement have missing values, thus for feature selection we consider only certain features in final dataframe, hence we choose simply ignore the null values.



time	0
latitude	0
longitude	0
depth	0
mag	0
magType	0
nst	2796
gap	2797
dmin	4199
rms	0
net	0
id	0
updated	0
place	0
type	0
horizontalError	3316
depthError	0
magError	2870
magNst	2807
status	0
locationSource	0
magSource	0
dtype: int64	

Fig 3.3 : Null Value Counts

4. FEATURE ENGINEERING

Feature engineering is the pre-processing step of machine learning, which is used to transform raw data into features that can be used for creating a predictive model using Machine learning or statistical Modelling.

Feature engineering in machine learning aims to improve the performance of models. In this topic, we will understand the details about feature engineering in Machine learning.

In machine learning, the performance of the model depends on data pre-processing and data handling. But if we create a model without pre-processing or data handling, then it may not give good accuracy. Whereas, if we apply feature engineering on the same model, then the accuracy of the model is enhanced. Hence, feature engineering in machine learning improves the model's performance.

4.1 Need of feature engineering:

- **Better features mean flexibility.**

The flexibility in features will enable you to select the less complex models. Because less complex models are faster to run, easier to understand and maintain, which is always desirable.

- **Better features mean simpler models.**

If we have good features, we can better represent the complete data and use it to best characterize the given problem.

- **Better features mean better results.**

To obtain better results, we must need to use better features.

- Apart from features in dataset we focus on, we have done some feature engineering based on some considerations as follows:
- Set rolling window size for future prediction based on past values with fixed window size in past.

- We are creating new features based on rolling window size on average depth and average magnitude.

The features that are created:

- depth_avg_22 - average depth of 22 days(approx. 3 weeks) rolling period
- depth_avg_15 - average depth of 15 days(approx. 2 weeks) rolling period
- depth_avg_7 - average depth of 7 days(1 weeks) rolling period
- mag_avg_22 - average magnitude of 22 days(approx. 3 weeks) rolling period
- mag_avg_15 - average depth of 15 days(approx. 2 weeks) rolling period
- mag_avg_7 - average depth of 7 days(1 weeks) rolling period
- mag_outcome - labels 0,1 based on the threshold0-
no earthquake
1-earthquake

4.2 Rolling Average

A rolling average, referred as moving average sometimes, it's a metric that calculates trends over short periods of time using a set of data. Specifically, it helps calculate trends when they might otherwise be difficult to detect

.

4.2.1 Why rolling average?

- Rolling average is a statistical method that can be useful in earthquake forecasting.
- It allows for the identification of trends and patterns in seismic activity.
- By taking the average of seismic activity over a specified period of time, researchers can identify whether seismic activity is increasing, decreasing, or remaining constant, which can be useful in predicting future earthquakes.

For example:

Let's say that a seismologist is studying seismic activity in a particular region over the course of several months. They decide to use a rolling average to identify any trends in seismic activity over time.

They choose a time period of 7 days and calculate the average seismic activity for each 7-day period. They then plot this data on a graph to visualize any trends.

If the rolling average shows an increase in seismic activity over time, this could indicate that an earthquake is more likely to occur in the near future.

Conversely, if the rolling average shows a decrease in seismic activity over time, this could indicate that the risk of an earthquake is decreasing.

4.2.2 Steps to calculate rolling average:

Rolling average = sum of data over time / time period

- Determine the time period
- Add the earliest totals
- Divide the total by your time period
- Calculate the average for your next rolling period
- Continue the formula for each rolling period
- Complete the formula regularly

5. TRAINING PHASE

Models used are :

Boosting Algorithms:

- Adaboost classifier with DecisionTree.
- Adaboost classifier with RandomForest.
- XGBoost.

Hyperparameter Tuning:

- GridSearchCV

5.1 Why Boosting Algorithms ?

Boosting algorithms such as XGBoost, AdaBoost etc. are often good choices for addressing class imbalance in binary classification problems. The reason is that these algorithms work by iteratively training weak models on the data, where each subsequent model tries to correct the mistakes made by the previous models.

By doing so, boosting algorithms can improve the overall performance on imbalanced datasets.

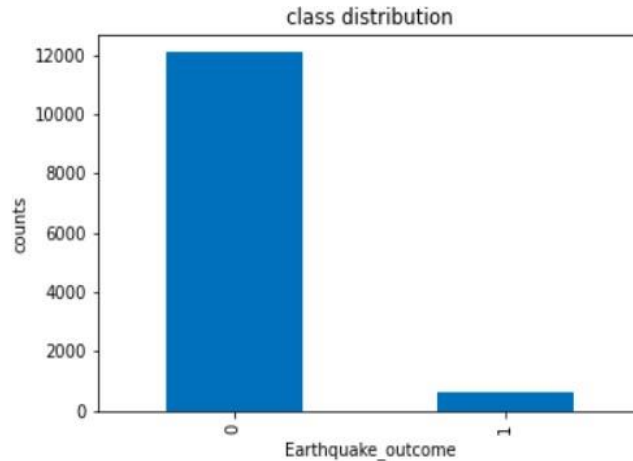


Fig:3.4 Class imbalance

5.1.1 Class Imbalance:

Class imbalance is a situation that occurs when the distribution of classes in a classification problem is not uniform, i.e., some classes have significantly fewer instances than others. Also it refers to the scenario where one class has a disproportionately large number of instances compared to other classes.

Class imbalance can have several effects on machine learning algorithms:

- **Biased models:** In case of class imbalance, a model may become biased towards the majority class, which leads to incorrect predictions for the minority class.
- **Poor performance on the minority class:** As the model is biased towards the majority class, it may have poor performance on the minority class. This can lead to lower recall and precision values for the minority class.

- **Overfitting:** In some cases, the model may overfit on the majority class due to the lack of samples for the minority class, which may lead to poor generalization.
- **Misclassification costs:** In some applications, misclassification costs may be higher for one class than the other. In this case, misclassifying the minority class can lead to a higher cost.
- **Inaccurate evaluation:** Inaccurate evaluation can occur when the training and testing sets have a different class distribution. This can lead to misleading performance metrics such as accuracy.
- **Overall,** class imbalance can lead to biased models, poor performance on the minority class, and inaccurate evaluation of the model. It is important to address class imbalance to ensure the model's performance is not affected by the skewed class distribution.

5.2 AdaBoost

AdaBoost, also called Adaptive Boosting, is a technique in Machine Learning used as an Ensemble Method. The most common estimator used with AdaBoost is decision trees with one level which means Decision trees with only 1 split. These trees are also called **Decision Stumps**.

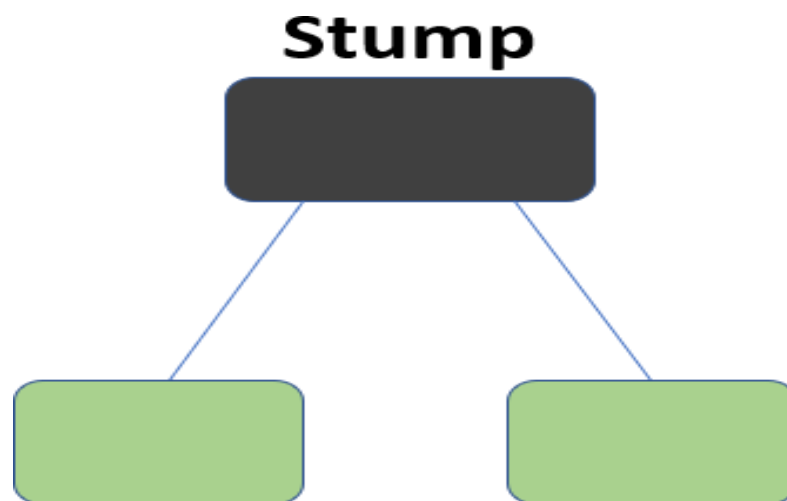


Fig 3.5 : Stump of the tree

What this algorithm does is that it builds a model and gives equal weights to all the data points. It then assigns higher weights to points that are wrongly classified. Now all the points with higher weights are given more importance in the next model. It will keep

training models until and unless a lower error is received.

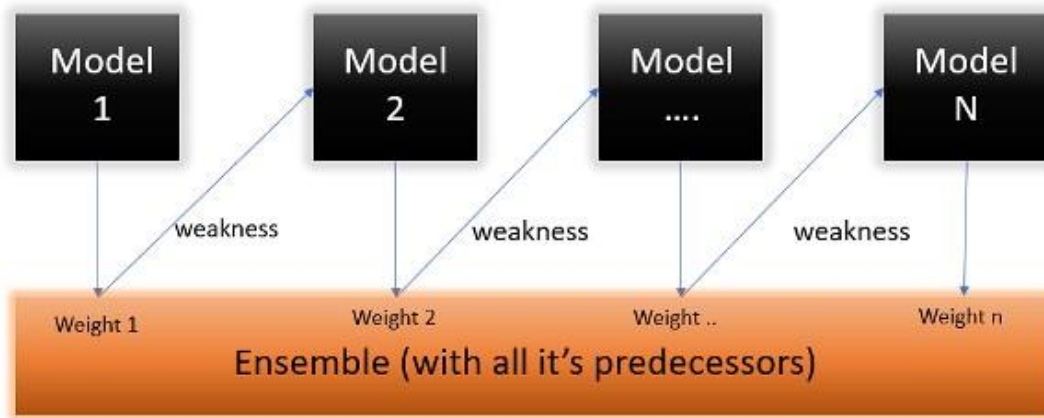


Fig 3.6 : Adaboost model

AdaBoost is used to boost the performance of decision trees on binary classification problems. It iteratively corrects the mistakes of the weak classifier and improves accuracy by combining weak learners. It is best used with weak learners.

5.2.1 Idea behind Ada Boost

AdaBoost combines a lot of “weak learners”(stumps) to make classifications.

1. Build model
2. assign higher weights to miss-classified points
3. Build new model.
4. Repeat 1,2 until we get the correct classification.

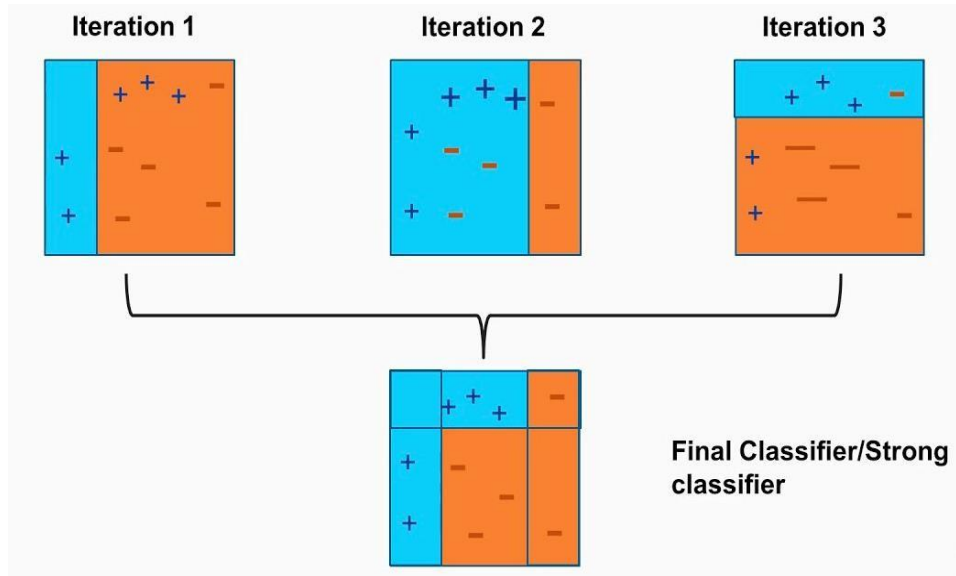


Fig 3.7 : Iterations of AdaBoost Model

5.2.2 How AdaBoost works:

Step 1: Assign a sample weight for each sample.

The formula to calculate the sample weights is:

$$w(x_i, y_i) = \frac{1}{N}, \quad i = 1, 2, \dots, n$$

Where N is the total number of data points.

Step 2: Calculate the Gini Impurity for each variable.

Gini impurity = 1 - (the probability of true)² - (the probability of false)²

Step 3: Calculate the Amount of Say for the stump that was created.

$$\frac{1}{2} \log \frac{1 - \text{Total Error}}{\text{Total Error}}$$

Step 4: Calculate the new sample weights for the next stump

Step 5: Create a bootstrapped dataset with the odds of each sample being chosen based on their new sample weights.

Step 6: Repeat the process n number of times.

So simply higher points are assigned to the data points which are miss-classified or incorrectly predicted.

5.3. Building Adaboost Model

AdaBoost uses Decision Tree Classifier as default Classifier.

"The most important parameters are `base_estimator`, `n_estimators`, and `learning_rate`."

- **base_estimator:** It is a weak learner used to train the model. It uses `DecisionTreeClassifier` as default weak learner for training purpose. You can also specify different machine learning algorithms.
- **n_estimators:** Number of weak learners to train iteratively.
- **learning_rate:** It contributes to the weights of weak learners. It uses 1 as a default value.

Here we have used Decision Tree Classifier and Random Forest as a base estimator. You can use any ML learner as base estimator if it accepts sample weight .

5.3.1 AdaBoost classifier with Decision tree :

A decision tree classifier is a machine learning algorithm used for classification tasks. It builds a tree-like model of decisions and their possible consequences, where each internal node represents a test on a feature, each branch represents the outcome of the test, and each leaf node represents a class label. The decision tree classifier works by recursively partitioning the feature space based on the values of the features. The algorithm starts at the root node of the tree and selects the feature that best separates the data based on some criterion, such as information gain or Gini impurity. It then splits the data into subsets based on the values of the selected feature and repeats the process on each subset until a stopping criterion is met.

Once the decision tree is built, it can be used to classify new instances by traversing the tree from the root node down to a leaf node, based on the values of the features of the instance. The class label associated with the leaf node is then assigned to the instance.

Decision tree classifiers have several advantages, including their interpretability, fast training time, and ability to handle both categorical and continuous data. However, they can be prone to overfitting, especially if the tree is deep or if the dataset is noisy or imbalanced. To address these issues, techniques such as pruning, ensemble methods like random forests or gradient boosting, or using alternative classification algorithms may be used.

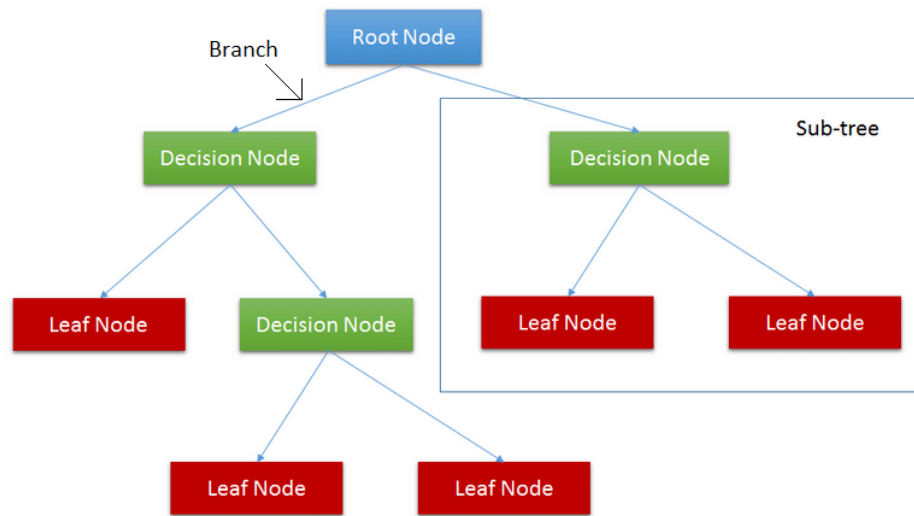


Fig 3.8 : Decision tree classifier

5.3.2 AdaBoost classifier with Random Forest :

Random forest is a machine learning algorithm that belongs to the family of ensemble methods. It is widely used for classification and regression tasks. The algorithm works by constructing a large number of decision trees and combining their predictions to make a final prediction. Each decision tree is trained on a random subset of the data, and a random subset of the features is selected at each node of the tree.

The key idea behind the random forest algorithm is to reduce overfitting by introducing randomness into the model. By constructing many decision trees with different subsets of data and features, random forest can reduce the variance of the model and improve its generalization performance.

To make a prediction with random forest, the algorithm first constructs a set of

decision trees. Each tree predicts the class label or the continuous value of the target variable based on the features of the input data.

The final prediction is then made by aggregating the predictions of all the trees. In classification tasks, the class with the most votes is chosen as the final prediction, while in regression tasks, the average of the predictions is taken.

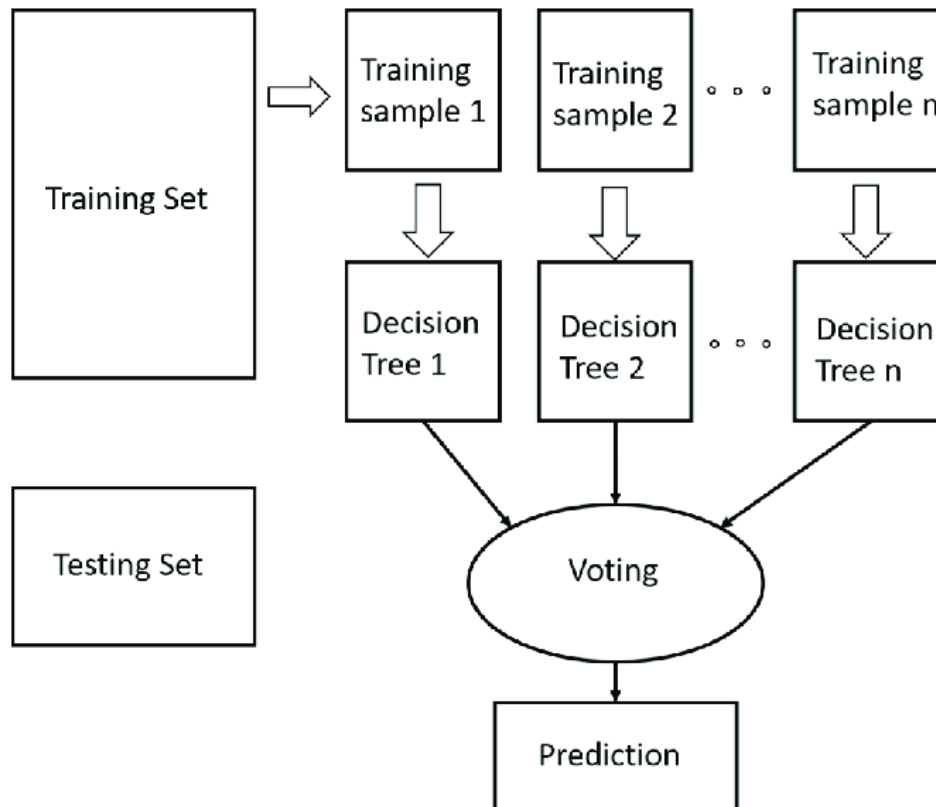


Fig 3.9 : Random Forest Classifier

Pros:

- AdaBoost is easy to implement.
- It iteratively corrects the mistakes of the weak classifier and improves accuracy by combining weak learners.
- You can use many base classifiers with AdaBoost. AdaBoost is not prone to overfitting.

Cons:

- AdaBoost is sensitive to noise data.
- It is highly affected by outliers because it tries to fit each point perfectly. AdaBoost is slower compared to XGBoost.

5.4 XGBoost

XGBoost stands for “Extreme Gradient Boosting”. XGBoost is a popular and efficient open-source implementation of the gradient boosted trees algorithm. Gradient boosting is a supervised learning algorithm, which attempts to accurately predict a target variable by combining the estimates of a set of simpler, weaker models. When using gradient boosting for regression, the weak learners are regression trees, and each regression tree maps an input data point to one of its leaves that contains a continuous score.

The training proceeds iteratively, adding new trees that predict the residuals or errors of prior trees that are then combined with previous trees to make the final prediction. It's called gradient boosting because it uses a gradient descent algorithm to minimize the loss when adding new models.

In particular, XGBoost is a popular boosting algorithm that has shown strong performance on a wide range of classification problems, including those with imbalanced datasets.

XGBoost has several features that make it well-suited for addressing class imbalance, such as the ability to assign higher weights to the minority class instances. XGBoost have specific parameters designed to handle imbalanced data, such as `scale_pos_weight`.

After preprocessing the data we can use XGBoost to train a model that predicts the occurrence of the earthquake. The input features are fed into the XGBoost model, which consists of an ensemble of weak decision trees.

The general process in XGBoost model is:

1. Will create a model.
2. Calculate the errors based on the previous model(Residuals).
3. Build a model to predict those errors.
4. Add last model to the ensemble of models.
5. We will repeat step 2,3,4 again upto the considered iterations.

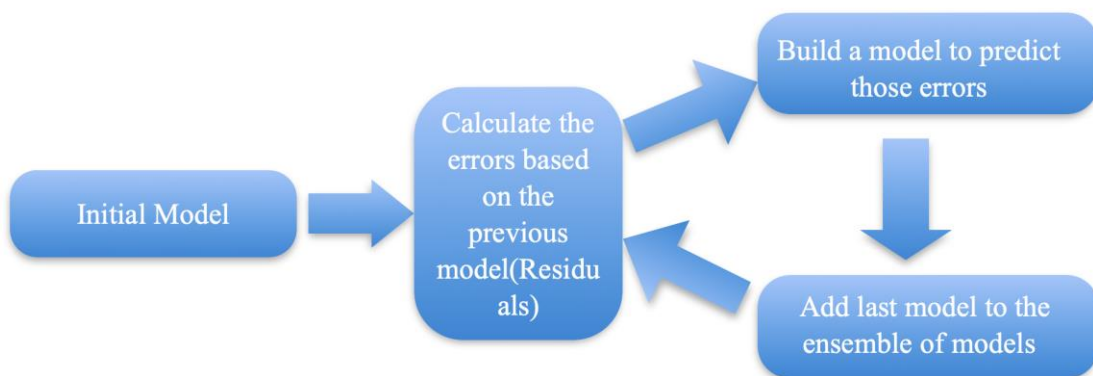


Fig 3.10 : XgBoost Model

5.5 Hyperparameter tuning

Hyperparameters are the parameters that are not learned from the training data but are set prior to the training process and are kept constant during the training process. Therefore, to find the optimal set of hyperparameters, one needs to perform multiple training iterations with different hyperparameters and evaluate the performance of the model. In our project we are using GridSearchCV. GridSearchCV is a hyperparameter tuning technique used to exhaustively search over a specified hyperparameter space for the best combination of hyperparameters that optimize a specified evaluation metric. The algorithm performs a cross-validation process on each hyperparameter combination to estimate the model performance and select the best combination that results in the highest evaluation metric.

For all the above algorithms,

- **max_depth** hyperparameter along with n_estimator was important as this indicates how deep the tree can be. The deeper the tree, the more splits it has and it captures more information about the data due earthquake data being only for past 30 days and features such as rolling window time period of magnitude.
- **max_features** hyperparameter is used since it ensures how many features to take in account for classification. Due to features such as magnitude and depth of quake for 22,15,and 7 days, this hyperparameter takes care of how many to pay attention to. GridSearchCV will take care of what features to take depending on $\sqrt{\text{num_features}}$, $\log(\text{num_features})$, $\text{auto}(\text{num_features})$.
- **n_estimators** It controls the number of weak learners or base estimators to include in the ensemble. It determines the number of iterations of the adaboost algorithm. The increase in the number of estimators can improve the performance of the model , up to certain point

CHAPTER 4

EXPERIMENTATION

1. EXPERIMENTATION

This chapter explains what we did and how we did this model, allowing readers to evaluate the reliability and validity of the experiment. In this chapter we are going to discuss the experimentation for the proposed model. We are going to explain the experimental setup required and various packages and libraries needed to be installed in the section 4.1. The datasets that are used in performing the experimentation is being explained in the section 4.2

The execution of our design process is as shown below :

1. EXPERIMENTAL SETUP

This project mainly uses the Machine Learning Techniques. It has been developed in Python language. Here all the required libraries and packages are installed.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
from sklearn.model_selection import train_test_split
from sqlalchemy import create_engine
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV
```

Fig. 4.1: Importing libraries.

- **Matplotlib:**

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

- **Numpy:**

NumPy is a general-purpose array-processing package. It provides a high performance multidimensional array object, and tools for working with these arrays. It is the fundamental package for scientific computing with Python.

- **Pandas:**

Pandas is an open source Python package that is most widely used for data science/data analysis and machine learning tasks. It is built on top of another package named Numpy, which provides support for multi-dimensional arrays. Text preprocessing is a method to clean the text data and make it ready to feed data to the model. Text data contains noise in various forms like emojis, numbers, punctuations, special characters, links.

- **SkLearn:**

The sklearn library contains a lot of efficient tools for machine learning and statistical modelling including classification, regression, clustering and dimensionality reduction.

1.1 SYSTEM REQUIREMENTS

System requirements are the system services and constraints the requirements that are generated during engineering process. The requirements mentioned are the software used in the project.

Operating System: Windows 8.

Tools: Google Colab, Jupyter Notebook.

1.2 HARDWARE REQUIREMENTS

Hardware Requirement is the system capacity levels in the process of engineering.

System: A stand-alone computer (i3 8th gen, 4gbRAM or higher).

Speed: 2.30GHz

RAM : 4.00GB System

Type: 64-bit Operating System

1.3 SOFTWARE REQUIREMENTS

click==7.1.2

Flask==1.1.2

gunicorn==20.0.4

itsdangerous==1.1.0

Jinja2==2.11.2

joblib==0.16.0

MarkupSafe==1.1.1

numpy==1.19.1

pandas==1.1.0

python-dateutil==2.8.1

pytz==2020.1

scikit-learn==0.23.1

scipy==1.5.2

six==1.15.0

sklearn==0.0

SQLAlchemy==1.3.18

threadpoolctl==2.1.0

Werkzeug==1.0.1

xgboost==1.1.1

2.DATASET USED

- We have collected the earthquake dataset from USGS(United States GeologicalSurvey).
- It is a real time data that updates every minute on.
- <https://earthquake.usgs.gov/earthquakes/feed/v1.0/csv.php> .
- Our dataset has 22 features.

all month (1)																					
time	latitude	longitude	depth	mag	magType	net	gap	dist	ms	net_id	updated	place	type	horizontalError	depthError	magError	mag1st	status	locationSource	magSource	
2023-02-08T06:57:27.030Z	38.883151	-122.6161887		1.78	1.56	nd	21	51	0.000001	0.01	nc	nc70843476	earthquake	0.18	0.44	0.08	20	automatic	nc	nc	
2023-02-08T06:59:09.430Z	36.9614983	-121.6100096	7.34	1.54	nd	8	72	0.00158	0.00	nc	nc70843459	earthquake	0.47	1.06	0.33	10	automatic	nc	nc		
2023-02-08T06:59:23.102Z	67.8838	-150.5006	86.2	1.7	nd				0.26	ak	ak0231wmp01	earthquake		0.6				automatic	ak	ak	
2023-02-08T06:59:27.13102Z	36.9613547	-121.588069	5.03	1.54	nd	11	76	0.00883	0.00	nc	nc70843461	earthquake	0.38	0.90	0.37	10	automatic	nc	nc		
2023-02-08T06:59:33.18302Z	19.1206601005547	-115.402328491211	21.710000102939	2.1400001	nd	42	180	0.119999997	nv	nv70221252	2023-02-08T06:59:41.840Z	10 km SE of Pihaike, Hawaii	earthquake	0.57	0.00000007	1.2999999	22	automatic	nv	nv	
2023-02-08T06:16:32.372Z	39.8792	-153.2487	120.5	2.4	nd				0.56	ak	ak0231twm0v	earthquake		0.5				automatic	ak	ak	
2023-02-08T06:05:03.050Z	33.025	-116.4531667	5.88	1.47	nd	57	54	0.02938	0.21	ci	ci04112180	earthquake	0.2	0.48	0.21	25	automatic	ci	ci		
2023-02-08T06:01:10.380Z	33.978833	-116.4756667	6.36	0.8	nd	20	105	0.06789	0.16	ci	ci04112184	earthquake	0.34	0.72	0.111	15	automatic	ci	ci		
2023-02-08T06:46:09.260Z	38.8271515	-122.7971969	2.0	1.05	nd	20	36	0.01078	0.02	nc	nc70843446	earthquake	0.25	0.50	0.19	20	automatic	nc	nc		
2023-02-08T06:46:59.240Z	19.16883877025	-115.459494973047	31.261000258819	1.8400000	nd	25	158	0.109999999	nv	nv70221257	2023-02-08T06:47:24.940Z	4 km SSW of Pihaike, Hawaii	earthquake	0.69	1.0499999	0.010000025	3	automatic	nv	nv	
2023-02-08T06:36:59.230Z	38.804231	-122.590216	1.74	0.28	nd	11	89	0.00191	0.01	nc	nc70843441	earthquake	0.38	0.91	0.18	11	automatic	nc	nc		
2023-02-08T06:11:08.330Z	38.799326	-122.529996	0.19	0.87	nd	7	100	0.01662	0.01	nc	nc70843456	earthquake	0.8	3.87	0.25	8	automatic	nc	nc		
2023-02-08T06:09:31.880Z	33.075	-116.716	2.19	0.42	nd	10	138	0.02169	0.15	ci	ci04112176	earthquake	0.74	0.37	0.195	9	automatic	ci	ci		
2023-02-08T06:06:27.880Z	38.840886	-122.676033	2.96	0.76	nd	13	100	0.006065	0.02	nc	nc70843451	earthquake	0.56	0.63	0.19	17	automatic	nc	nc		
2023-02-08T07:01:09.807Z	39.0033	-152.6905	61.3	2	nd				1.03	ak	ak0231twm0u	earthquake		1.4				automatic	ak	ak	
2023-02-08T07:00:14.330Z	19.133342	-115.4806	155.3743215332	25.332000065455	2.11888889	nd	40	184	0.109999999	nv	nv70221212	2023-02-08T07:02:34.560Z	15 km SW of Pihaike, Hawaii	earthquake	0.66	1.25	0.700000019	17	automatic	nv	nv
2023-02-08T07:48:02.134Z	37.8983	36.4734	10	4.9	nd	70	24	0.983	0.86	us	us0000mg1	earthquake	4.11	1.807	0.082	47	reviewed	us	us		
2023-02-08T07:47:20.910Z	38.7804885	-122.7549984	0.88	0.75	nd	10	102	0.006781	0.01	nc	nc70843426	earthquake	0.29	0.48	0.18	9	automatic	nc	nc		
2023-02-08T07:45:00.738Z	59.8076	-152.78237	61.4	1.9	nd				0.62	ak	ak0231twm7n	earthquake		0.7				automatic	ak	ak	
2023-02-08T07:36:10.807Z	62.8074	-151.3087	114.7	1.6	nd				0.12	ak	ak0231twm0b0	earthquake		1.3				automatic	ak	ak	
2023-02-08T07:34:39.818Z	62.0453	-149.6879	58.2	1.6	nd				0.22	ak	ak0231twm0p	earthquake		0.9				automatic	ak	ak	
2023-02-08T07:30:23.130Z	38.7944984	-122.730235	0.59	2.16	nd	37	69	0.01302	0.04	nc	nc70843421	earthquake	0.18	0.37	0.19	38	automatic	nc	nc		
2023-02-08T07:31:47.030Z	34.8035	-118.5691887	5.67	1.38	nd	14	115	0.00107	0.3	ci	ci04112186	earthquake	1.11	0.65	0.511	21	automatic	ci	ci		
2023-02-08T07:11:02.730Z	18.151	-87.889	82.77	3.43	nd	11	304	0.003	0.18	pr	pr71856568	earthquake	1.39	1.08	0.89329968959113	6	reviewed	pr	pr		
2023-02-08T06:59:05.030Z	19.2639991790564	-115.41543794805	32.2799997762090	2.83	nd	46	120	0.129999995	nv	nv70221177	2023-02-08T06:59:03.040Z	44 km W of Tynes, Alaska	earthquake	0.49	0.59999994		3.11	28	automatic	nv	nv
2023-02-08T06:58:06.830Z	38.799992	-122.6180334	10.91	0.77	nd	6	213	0.01189	0.02	nc	nc70843417	earthquake	3.03	7.56	0.4	8	automatic	nc	nc		
2023-02-08T06:56:01.947Z	47.8888	-145.5589	53.835	4.4	nd	61	134	1.887	0.41	us	us0000ym6	earthquake	8.87	7.583	0.079	56	reviewed	us	us		
2023-02-08T06:52:07.290Z	38.791182	-122.710007	3.44	0.78	nd	7	96	0.01462	0.01	nc	nc70843406	earthquake	0.45	2.83	0.00	8	automatic	nc	nc		
2023-02-08T06:28:36.250Z	5.6244	-82.8747	10	4.6	nd	34	224	2.841	0.43	us	us0000ym	earthquake	8.63	1.946	0.133	19	reviewed	us	us		
2023-02-08T06:19:47.871Z	61.5355	-141.943	4.8	1.5	nd				0.47	ak	ak0231twm0k	earthquake		0.6				automatic	ak	ak	
2023-02-08T06:16:49.877Z	62.6073	-151.0868	79.1	2.1	nd				0.49	ak	ak0231twm0p	earthquake		0.5				automatic	ak	ak	
2023-02-08T06:15:11.840Z	38.788347	-122.7289963	0.59	0.95	nd	6	130	0.01218	0.02	nc	nc70843401	earthquake	0.35	2.87	0.22	7	automatic	nc	nc		
2023-02-08T06:11:04.260Z	61.1348	-151.9442	95.5	2.3	nd				0.40	ak	ak0231twm0n	earthquake		0.9				automatic	ak	ak	
2023-02-08T06:10:05.440Z	38.794992	-122.728677	1.03	1.12	nd	9	103	0.01297	0.02	nc	nc70843396	earthquake	0.28	0.79	0.29	9	automatic	nc	nc		
2023-02-08T06:09:01.807Z	36.3073	36.1303	10	4.4	nd	39	186	1.185	0.59	us	us0000ym	earthquake	4.45	1.965	0.11	24	reviewed	us	us		

Fig. 4.2: Dataset

Following are the features of the Dataset:

- time ----- Time when the event occurred (milliseconds).
- latitude----- Decimal degrees latitude. Negative values for southern latitudes.
- longitude ----- Decimal degrees longitude. Negative values for western longitudes.
- depth -----Depth of the event in kilometers.
- mag ----- Magnitude of event occurred.
- magType ----- The method or algorithm used to calculate the preferred magnitude
- nst -----The total number of seismic stations used to determine earthquake location.
- gap ----- The largest azimuthal gap between azimuthally adjacent stations (in degrees).
- dmin -----Horizontal distance from the epicenter to the nearest station (in degrees).
- rms -----The root-mean-square (RMS) travel time residual, in sec, using all weights.
- net -----The ID of a data source contributor for event occurred.
- id ----- A unique identifier for the event.
- types ----- A comma-separated list of product types associated to this event.
- place ----- named geographic region near to the event.
- type ----- Type of seismic event.
- locationSource ----- The network that originally authored the reported location of this event.
- magSource ----- Network that originally authored the reported magnitude for this event.
- horizontalError ----- Uncertainty of reported location of the event in kilometers.
- depthError----- The depth error, three principal errors on a vertical line.
- magError -----Uncertainty of reported magnitude of the event.

- **magNst** ----- The total number of seismic stations to calculate the magnitude of earthquake.
- **status** ----- Indicates whether the event has been reviewed by a human.

2.1 LOADING THE DATASET:

```
df=pd.read_csv('all_month.csv')
```

Fig. 4.3: Loading the dataset

all_month (1)																							
time	latitude	longitude	depth	mag	magType	nst	gap	dmn	rms	net	id	updated	place	type	horizontalError	depthError	magError	magNst	status	locationSource	magSource		
2023-02-08T09:57:27.0302	38.831501	-122.8161697		1.78	1.06	md	21	51	0.003931	0.01	nc	nc73843476	2023-02-08T09:58:04.0162	8km NW of The Geysers, CA	earthquake	0.18	0.44	0.08	20	automatic	nc	nc	
2023-02-08T09:33:59.4302	36.9614983	-121.6100006		7.34	1.24	md	8	72	0.05158	0.06	nc	nc73843456	2023-02-08T09:46:10.8632	6km SW of Gilroy, CA	earthquake	0.47	1.06	0.33	10	automatic	nc	nc	
2023-02-08T09:29:23.8102	62.8838	-150.5088	96.2	1.7	ml					0.26	ak	ak0231enfbt1	2023-02-08T09:31:24.0332	45 km NNE of Petersville, Alaska	earthquake		0.6			automatic	ak	ak	
2023-02-08T09:27:15.6102	36.9615342	-121.599999		5.33	1.24	md			11.79	0.0583	0.09	nc	nc73843461	2023-02-08T09:50:10.8012	6km SSW of Gilroy, CA	earthquake	0.38	0.93	0.37	10	automatic	nc	nc
2023-02-08T09:25:18.2302	19.1356666855547	-185.402358491211	31.7000007620399	2.1400001	md	42	180		0.119999997	hv	hv73321262	2023-02-08T09:26:41.8402	10 km SE of Papeete, Hawaii	earthquake	0.97	0.930000007	1.29999999	22	automatic	hv	hv		
2023-02-08T09:16:32.3722	59.9782	-153.2467	120.5	2.4	ml					0.56	ak	ak0231enfbt1	2023-02-08T09:19:13.3202	52 km ENE of Pedro Bay, Alaska	earthquake		0.5			automatic	ak	ak	
2023-02-08T09:05:03.0502	33.505	-116.4531667		5.88	1.47	ml	57	54	0.02938	0.21	ci	ci04121192	2023-02-08T09:15:54.0402	21km ESE of Anza, CA	earthquake	0.2	0.48	0.21	25	automatic	ci	ci	
2023-02-08T09:01:10.6302	33.978833	-116.4759667		9.35	0.8	ml	20	105	0.06739	0.16	ci	ci04121184	2023-02-08T09:15:52.0732	3km NE of Desert Hot Springs, CA	earthquake	0.34	0.72	0.111	15	automatic	ci	ci	
2023-02-08T08:48:06.2002	38.8271675	-122.7971649		2.6	1.05	md	20	36	0.01076	0.02	nc	nc73843446	2023-02-08T09:45:18.5302	6km W of Cobb, CA	earthquake	0.25	0.55	0.19	20	automatic	nc	nc	
2023-02-08T08:44:06.3402	38.824501	-122.805016		1.74	0.28	md	11	99	0.001191	0.01	nc	nc73843441	2023-02-08T08:33:16.4732	7km NW of The Geysers, CA	earthquake	0.69	1.04999995	0.810000026	3	automatic	hv	hv	
2023-02-08T08:36:29.5302	38.824501	-122.805016		1.74	0.28	md	11	99	0.001191	0.01	nc	nc73843441	2023-02-08T08:33:16.4732	7km NW of The Geysers, CA	earthquake	0.38	0.81	0.18	11	automatic	nc	nc	
2023-02-08T08:11:58.3302	38.7958336	-122.7229996		0.79	0.97	md	7	100	0.01653	0.01	nc	nc73843436	2023-02-08T09:08:16.3642	3km S of Cobb, CA	earthquake	0.6	3.87	0.25	8	automatic	nc	nc	
2023-02-08T08:06:31.3502	33.5775	-116.716		2.19	0.42	ml	10	138	0.02149	0.15	ci	ci04121176	2023-02-08T08:10:05.1902	5km WNW of Anza, CA	earthquake	0.74	0.37	0.195	9	automatic	ci	ci	
2023-02-08T08:06:27.6602	38.8409999	-122.876503		2.26	0.78	md	13	120	0.008265	0.02	nc	nc73843451	2023-02-08T08:59:17.2942	13km NW of The Geysers, CA	earthquake	0.55	0.63	0.19	17	automatic	nc	nc	
2023-02-08T08:06:27.6602	38.8409999	-122.876503		2.26	0.78	md	13	120	0.008265	0.02	nc	nc73843451	2023-02-08T08:59:17.2942	13km NW of The Geysers, CA	earthquake	0.55	0.63	0.19	17	automatic	nc	nc	
2023-02-08T07:50:14.9302	19.1353340149026	-185.37483215332	29.530000686405	2.11999999	md	40	184		0.159999996	hv	hv73321212	2023-02-08T07:53:24.5802	13 km SE of Pāhala, Hawaii	earthquake	0.66	1.25	0.730000019	17	automatic	hv	hv		
2023-02-08T07:48:32.1342	37.9993	36.4734		10	4.9	mb	70	24	0.983	0.96	us	us000ympt1	2023-02-08T09:20:27.0732	3 km SW of Gökseur, Turkey	earthquake	4.11	1.857	0.082	47	reviewed	us	us	
2023-02-08T07:47:20.5102	38.7804985	-122.7649994		0.86	0.75	md	10	102	0.008781	0.01	nc	nc73843426	2023-02-08T08:37:29.1052	16m WNW of The Geysers, CA	earthquake	0.29	0.46	0.18	9	automatic	nc	nc	
2023-02-08T07:45:00.7282	59.9076	-152.2637		61.4	1.9	ml				0.62	ak	ak0231enfbt1	2023-02-08T07:47:46.4052	28 km WNW of Anchor Point, Alaska	earthquake		0.7			automatic	ak	ak	
2023-02-08T07:36:10.8072	62.9274	-151.3087		114.7	1.6	ml				0.12	ak	ak0231enfbt1	2023-02-08T07:38:36.4832	55 km WNW of Petersville, Alaska	earthquake		1.3			automatic	ak	ak	
2023-02-08T07:34:36.8182	62.0455	-149.6879		58.2	1.6	ml				0.22	ak	ak0231enfbt1	2023-02-08T07:36:12.5952	15 km SSE of Sustut North, Alaska	earthquake		0.9			automatic	ak	ak	
2023-02-08T07:33:23.1302	38.7944984	-122.7283325		0.59	2.16	md	37	69	0.01302	0.04	nc	nc73843421	2023-02-08T09:46:10.5472	3km NE of The Geysers, CA	earthquake	0.16	0.37	0.19	38	automatic	nc	nc	
2023-02-08T07:21:47.6202	34.0009	-118.4481967		2.62	1.28	ml	14	175	0.02022	0.3	ci	ci04121168	2023-02-08T07:25:29.6942	14km WNW of Matlacha, CA	earthquake	1.11	0.89	0.211	21	automatic	ci	ci	
2023-02-08T07:11:15.7102	18.161	-87.589		83.77	3.43	md	11	824	0.533	0.18	pr	pr73595658	2023-02-08T07:14:13.2002	60 km SE of Punta Cana, Dominican Republic	earthquake	1.96	1.08	0.03289885896113	6	reviewed	pr	pr	
2023-02-08T06:53:50.3902	19.2026995795054	-155.134576705805	32.2799987792693	2.83	ml	45	120		0.129999995	hv	hv73321177	2023-02-08T08:28:03.0402	9 km NE of Pāhala, Hawaii	earthquake	0.49	0.589999974	3.11	29	automatic	hv	hv		
2023-02-08T06:39:06.8302	38.7999992	-122.8188324		10.91	0.77	md	6	213	0.01199	0.02	nc	nc73843411	2023-02-08T07:41:32.7002	6km WNW of The Geysers, CA	earthquake	3.03	7.56	0.4	8	automatic	nc	nc	
2023-02-08T06:27:01.9472	42.9888	145.3592		53.335	4.4	mb	61	134	1.897	0.41	us	us000ymf1	2023-02-08T08:02:42.0402	41 km SSW of Nemuro, Japan	earthquake	8.87	7.583	0.072	56	reviewed	us	us	
2023-02-08T06:27:07.0902	38.7811682	-122.7210037		3.44	0.78	md	7	96	0.01462	0.01	nc	nc73843406	2023-02-08T07:29:45.6272	3km NW of Anderson Springs, CA	earthquake	0.45	2.83	0.05	8	automatic	nc	nc	
2023-02-08T06:26:36.2552	5.6244	-82.6747		10	4.6	mb	34	224	2.841	0.43	us	us000ymf1	2023-02-08T07:41:52.0402	267 km S of Burica, Panama	earthquake	8.63	1.946	0.133	19	reviewed	us	us	
2023-02-08T06:19:47.8712	61.3325	-141.943		4.8	1.3	ml				0.42	ak	ak0231enfbt1	2023-02-08T06:23:01.5402	53 km ESE of McCarthy, Alaska	earthquake		0.5			automatic	ak	ak	
2023-02-08T06:15:49.0772	62.6073	-151.0868		78.1	2.1	ml				0.49	ak	ak0231enfbt1	2023-02-08T06:18:15.9442	20 km NW of Petersville, Alaska	earthquake		0.5			automatic	ak	ak	
2023-02-08T06:13:11.8402	38.7938347	-122.7289963		-0.92	0.99	md	6	130	0.01218	0.07	nc	nc73843401	2023-02-08T06:30:11.2712	3km NE of The Geysers, CA	earthquake	0.35	2.87	0.22	7	automatic	nc	nc	
2023-02-08T06:11:54.3502	61.1348	-151.9452		95.6	2.3	ml				0.45	ak	ak0231enfbt1	2023-02-08T06:13:39.0402	24 km W of Tyonek, Alaska	earthquake		0.9			automatic	ak	ak	
2023-02-08T06:10:06.4402	38.7949982	-122.7296677		1.03	1.13	md	9	103	0.01297	0.02	nc	nc73843396	2023-02-08T09:30:10.4742	3km NE of The Geysers, CA	earthquake	0.26	0.79	0.26	9	automatic	nc	nc	
2023-02-08T06:09:01.8272	36.3573	36.1303		10	4.4	mb	39	186	1.188	0.59	us	us000ymf1	2023-02-08T07:09:30.3472	Turkey-Syria border region	earthquake	4.45	1.965	0.11	24	reviewed	us	us	
2023-02-08T06:08:32.8302	35.698	-117.560967		6.07	0.62	ml	8	149	0.1209	0.13	ci	ci04121160	2023-02-08T06:12:27.4502	3km NE of Ridgecrest, CA	earthquake	0.46	2.3	0.049	7	automatic	ci	ci	
2023-02-08T06:56:28.6102	38.7956657	-122.7304997		0.55	0.89	md	12	100	0.01333	0.03	nc	nc73843388	2023-02-08T05:58:03.1352	3km SSW of Cobb, CA	earthquake	0.2	0.9	0.15	9	automatic	nc	nc	
2023-02-08T06:56:23.1102	38.8491669	-122.7795029		1.23	1.12	md	15	154	0.008989	0.03	nc	nc73843381	2023-02-08T09:37:16.4932	6km WNW of Cobb, CA	earthquake	0.31	0.43	0.5	15	automatic	nc	nc	
2023-02-08T06:54:05.1802	38.7956657	-122.7275029		1.31	0.83	md	8	100	0.02565	0.03	nc	nc73843376	2023-02-08T05:56:43.0012	3km S of Cobb, CA	earthquake	0.29	1.35	0.11	8	automatic	nc	nc	
2023-02-08T06:53:10.0002	38.7956657	-122.7303314		1.14	1.06	md	7	139	0.01338	0.01	nc	nc73843371	2023-02-08T05:54:43.1252	3km SSW of Cobb, CA	earthquake	0.33	1.4	0.07	9	automatic	nc	nc	
2023-02-08T05:53:00.7402	38.8260002	-122.7953333		2.59	0.98	md	21	38	0.01104	0.01	nc	nc73843366	2023-02-08T05:54:37.9482	6km W of Cobb, CA	earthquake	0.22	0.6	0.27	22	automatic	nc	nc	

Fig:4.4:Dataset

In [4]: `df.head()`

Out[4]:

	time	latitude	longitude	depth	mag	magType	nst	gap	dmin	rms	...	updated	place	type	horizontalError	dep
0	2023-02-05T04:29:58.260Z	34.193833	-117.442500	3.45	1.65	ml	66.0	36.0	0.01254	0.24	...	2023-02-05T04:40:41.920Z	5km SW of Devore, CA	earthquake	0.24	
1	2023-02-05T04:07:02.510Z	47.700333	-120.135167	-0.33	0.87	ml	15.0	53.0	0.07089	0.13	...	2023-02-05T04:31:50.420Z	6 km ENE of Entiat, Washington	earthquake	0.23	
2	2023-02-05T04:00:15.928Z	58.609200	-153.820000	89.60	1.60	ml	NaN	NaN	NaN	0.20	...	2023-02-05T04:02:17.196Z	81 km NW of Aleneva, Alaska	earthquake	NaN	
3	2023-02-05T03:55:28.950Z	33.694167	-116.740333	16.93	0.70	ml	27.0	50.0	0.02763	0.15	...	2023-02-05T03:59:31.198Z	6km SSW of Idyllwild, CA	earthquake	0.32	
4	2023-02-05T03:52:49.381Z	62.824600	-150.755000	92.50	1.40	ml	NaN	NaN	NaN	0.59	...	2023-02-05T03:55:50.971Z	36 km N of Petersville, Alaska	earthquake	NaN	

5 rows x 22 columns

In [5]: `df.shape`

Out[5]: (9558, 22)

Fig. 4.5: Dataset head

3 DATA PREPROCESSING:

Null values Input to model from dataset has many important features to consider as time, latitude & longitude, depth of quake, magnitude, place, rest other features are error and non supporting features for classification, below shows the null value counts for some features and what to do with that.

8]: `df.isnull().sum()`

8]:

time	0
latitude	0
longitude	0
depth	0
mag	0
magType	0
nst	2796
gap	2797
dmin	4199
rms	0
net	0
id	0
updated	0
place	0
type	0
horizontalError	3316
depthError	0
magError	2870
magNst	2807
status	0
locationSource	0
magSource	0
dtype: int64	

Fig 4.6 : Null value counts

As there are lots of null values of certain features, but as part of prediction most of the features that address 'error' in measurement have missing values, thus for feature selection we consider only certain features in final dataframe, hence we choose simply ignore the null values.

4 FEATURE ENGINEERING

Apart from features in dataset we focus on, we have done some feature engineering based on some considerations as follows:

- Set rolling window size for future prediction based on past values with fixed window size in past.
- We are creating new features based on rolling window size on average depth and average magnitude.
- A final outcome 'mag_outcome' has been defined as target values and the output is considered as shifted values from set rolling window of past days eg: '7'.
- Finally for mag_outcome feature we created based on 7 days rolling window period in future as target, we have converted it to class as 1 or 0 based on magnitude outcome > 2.51
- **New features include :** avg_depth, magnitude_avg for 22,15,7 days rolling window period for training.

Out[31]:

	date	depth	mag	place	latitude	longitude	depth_avg_22	depth_avg_15	depth_avg_7	mag_avg_22	mag_avg_15	mag_avg_7	mag_outcome
6845	2023-01-17	6.5	1.1	California	39.010098	-119.814634	7.918182	7.633333	6.457143	1.213636	1.486667	1.657143	0.342857
6846	2023-01-17	8.3	1.5	California	39.010098	-119.814634	7.659091	7.880000	7.642857	1.268182	1.546667	1.600000	0.085714
6847	2023-01-18	9.1	0.2	California	39.010098	-119.814634	7.681818	8.293333	7.514286	1.263636	1.386667	1.257143	0.157143
6848	2023-01-18	9.0	0.0	California	39.010098	-119.814634	8.090909	8.373333	7.128571	1.263636	1.306667	1.042857	0.128571
6849	2023-01-18	6.6	0.3	California	39.010098	-119.814634	8.040909	8.200000	7.800000	1.268182	1.240000	0.857143	0.200000
6850	2023-01-18	7.3	0.1	California	39.010098	-119.814634	8.245455	7.740000	7.757143	1.190909	1.146667	0.671429	0.357143
6851	2023-01-19	5.2	0.6	California	39.010098	-119.814634	7.822727	7.600000	7.428571	1.172727	1.133333	0.542857	0.685714
6852	2023-01-19	0.0	-0.3	California	39.010098	-119.814634	7.272727	7.046667	6.500000	1.122727	1.046667	0.342857	0.757143
6853	2023-01-19	6.1	-0.3	California	39.010098	-119.814634	7.340909	6.453333	6.185714	1.081818	0.913333	0.085714	1.014286
6854	2023-01-19	4.4	0.7	California	39.010098	-119.814634	7.409091	6.746667	5.514286	0.995455	0.833333	0.157143	1.057143

Fig 4.7: New features

- After feature engineering and dealing with null values, the model has imbalance class distribution.

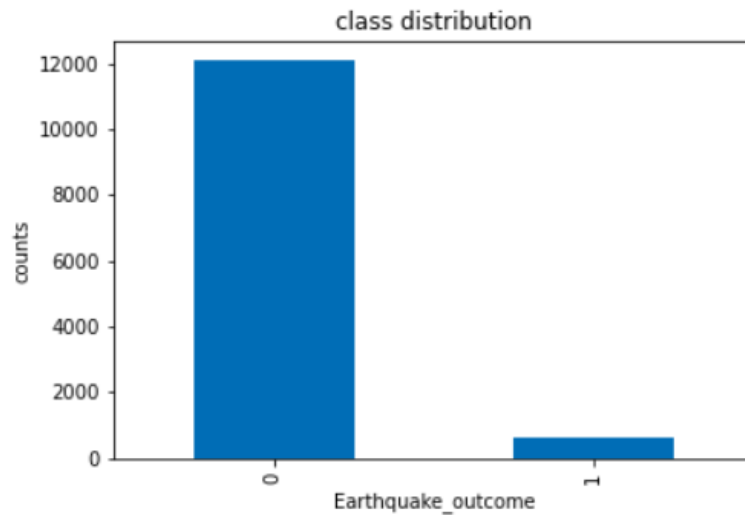


Fig 4.8 : Imbalance of class

- Lets analyse places with top 20 higher & lower number of magnitude mean:

```
In [17]: df.groupby(['place'])['mag'].mean().nsmallest(20).plot(kind='bar')
```

```
Out[17]: <AxesSubplot:xlabel='place'>
```

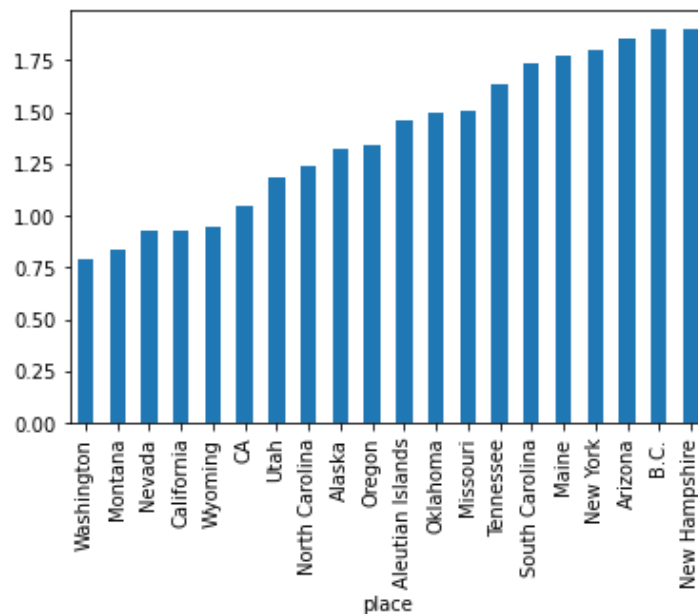


Fig:4.9 :Top 20 places where lowest magnitude mean quake experienced in past 30 days.

```
In [18]: df.groupby(['place'])['mag'].mean().nlargest(20).plot(kind='bar')
```

```
Out[18]: <AxesSubplot:xlabel='place'>
```

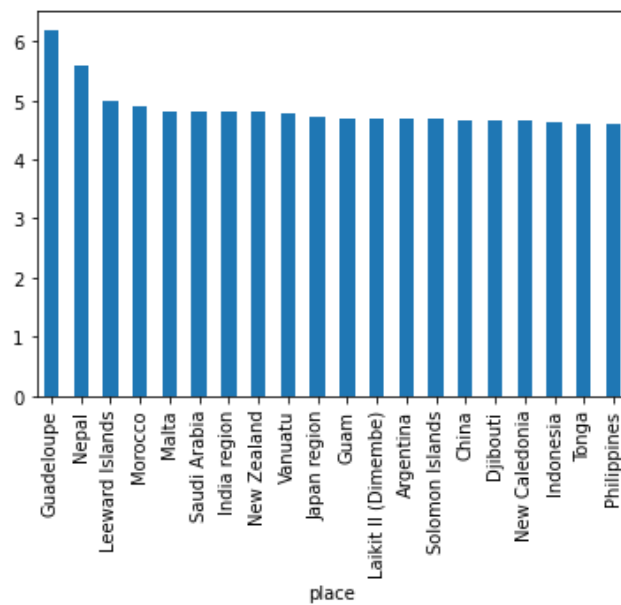


Fig:4.10:Top 20 places where highest magnitude mean quake experienced in past 30 days.

5 MODEL IMPLEMENTATION:

After preprocessing with removing null values, and feature engineering as discussed above, we performed Boosting algorithms for classification problem.

1. Adaboost classifier with estimator as DecisionTreeClassifier.
2. Adaboost classifier with estimator as RandomForestClassifier.
3. Xgboost algorithm.

For all the above algorithms,

1. Decision Tree Classifier:

`max_depth = [2,6,7]`, `n_estimators = [200,500,700]` and we used gridsearch CV for best estimator as nodes are expanded until all leaves are pure or until all leaves contain less than `min_samples_split = 2` samples which helps for classification with various types of features in dataset.

2. Random Forest Classifier:

Same parameters were used for randomforest as well to compare the algorithms used with gridsearchCV along with another hyperparameter `max_features = ['auto','sqrt','log2']` that will let select features based on $\log(\text{features})$, $\sqrt{\text{features}}$ etc.

3. Xgboost Classifier:

We did not use grid Search CV here since, it took more time to train, hence we tried `max_depth` same as above algorithms with best fit, i.e 6, `learning_rate=0.03` and `gbtree` as booster.

- Model selection was based on Evaluation on **roc_auc** score and recall and hyperparameter tuning.
- **max_depth** hyperparameter along with **n_estimator** was important as this indicates how deep the tree can be.
- The deeper the tree, the more splits it has and it captures more information about the data due earthquake data being only for past 30 days and features such as rolling window time period of magnitude.
- **max_features** hyperparameter is used since it ensures how many features to take in account for classification. Due to features such as magnitude and depth of quake for 22,15,and 7 days, this hyperparameter takes care of how many to pay attention to.
- GridSearchCV will take care of what features to take depending on $\sqrt{\text{num_features}}$, $\log(\text{num_features})$, `auto(num_features)`.

5.1 Implementation Of Adaboost Decision Tree Classifier:

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV

param_grid = {
    "base_estimator__max_depth" : [2,5,7],
    "n_estimators": [200, 400, 600]
}

# base estimator
tree = DecisionTreeClassifier(class_weight='balanced')

# adaboost with the tree as base estimator
# learning rate is arbitrarily set to 0.6,
ABC = AdaBoostClassifier(
    base_estimator=tree,
    learning_rate=0.6,
    algorithm="SAMME")
```

Fig: 4.11 Implementation of Adaboost Decision Tree Classifier

```
grid_search_ABC = GridSearchCV(ABC,
                                param_grid=param_grid,
                                scoring = 'roc_auc',
                                return_train_score=True,
                                verbose = 1)
```

Fig: 4.12 Hyperparameter Tuning Using GridSearchCV

```
grid_search_ABC.fit(X_train,y_train)
```

Fig:4.13:Fitting the model

```
pred_ABC=grid_search_ABC.predict(X_test)
```

Fig:4.14:Predicting the test data

5.2 Implementation Of Adaboost Random Forest Classifier:

```
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier

rfc = RandomForestClassifier(n_jobs=-1,max_features= 'sqrt',n_estimators=50, oob_score = True,class_weight='balanced')

param_grid = {
    'n_estimators': [200, 700],
    'max_features': ['auto', 'sqrt', 'log2']
}
# GridSearch of parameter tuning.
CV_rfc = GridSearchCV(estimator=rfc, param_grid=param_grid, cv= 5)
CV_rfc.fit(X_train, y_train)
pred=CV_rfc.predict(X_test)
```

Fig:4.15 Implementation of Adaboost Random Forest Classifier

5.3 Implementation Of XGBoost :

```
from xgboost import XGBClassifier
import matplotlib.pyplot as plt

# Calculate scale_pos_weight
neg, pos = np.bincount(y_train)
scale_pos_weight = neg / pos

# Create DMatrix
dtrain = xgb.DMatrix(X_train[features], label=y_train)
dtest = xgb.DMatrix(X_test[features], label=y_test)

# Set XGBoost parameters
params = {
    'objective': 'binary:logistic',
    'booster': 'gbtree',
    'eval_metric': 'auc',
    'max_depth': 6,
    'eta': 0.003,
    'scale_pos_weight': scale_pos_weight
}

# Train model
num_round = 5000
bst = xgb.train(params, dtrain, num_round)

# Predict on test data
preds = bst.predict(dtest)
```

Fig:4.16 Implementation Of XGBoost Classifier

CHAPTER 5

PERFORMANCE EVALUATION AND ANALYSIS

1.PERFORMANCE EVALUATION

We have used GridSearchCV for improving model and hyper parameter tuning on AdaBoost classifier with base estimators as DecisionTreeClassifier and RandomForestClassifier.

Using the same hyper parameters we trained XGBoost.

Our problem is about binary classification, where Earthquake occur = 1 and Earthquake not occur = 0 and with these prediction we try to locate co-ordinates corresponding to the predictions and display it on the google maps.

For this project we need to minimize or get less False negative predictions since we don't want our model to predict as 0 or no earthquake occurred at particular location when in reality it had actually happened.

As this is more dangerous than the prediction case in which prediction is true/1 or earthquake occurred but in reality it did not because its always **better safe than sorry!**

- Accuracy is not the metric to use when working with an imbalanced dataset. We have seen that it is misleading. There are metrics that have been designed to tell you a more truthful story when working with imbalanced classes. such as collect more data, change metrics, resampling data, cross-validation dataset etc. For the project I have considered the metrics for treating this imbalance nature with-
 1. Confusion Matrix: A breakdown of predictions into a table showing correct predictions (the diagonal) and the types of incorrect predictions made (what classes incorrect predictions were assigned).
 2. Recall: A measure of a classifiers completeness
 3. ROC Curves: Like precision and recall, accuracy is divided into sensitivity and specificity and models can be chosen based on the balance thresholds of these values.
- Moreover the reason for choosing this metrics not only helps me improve class imbalance comfirmation bias but also due to my nature of problem to be solved of earthquake prediction False negative must be penalized more.

```

from sklearn.metrics import roc_curve
from sklearn.metrics import auc
from sklearn.metrics import roc_auc_score
from sklearn.metrics import recall_score
from sklearn.metrics import confusion_matrix

```

Fig 4.9 : Importing necessary metrics.

2.RESULT ANALYSIS:

2.1 Adaboost Decision Tree Classifier:

1. With **adaboost decision tree classifier** and hyper parameter tuning, we get area under curve (score) = 0.939
2. Higher the auc score, better is the model since it is better at distinguishing positive and negative classes.
3. From **confusion matrix**, False negative = 8 and Recall score = 0.882.

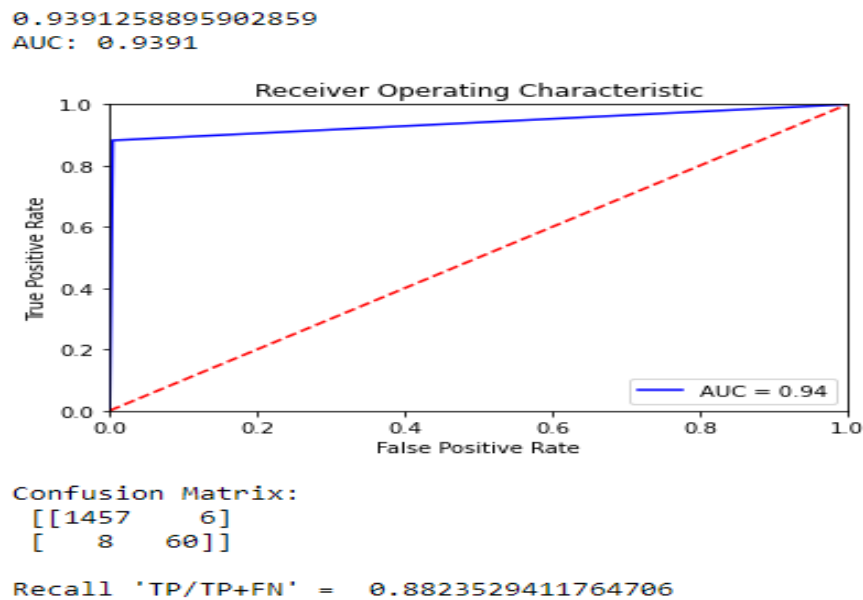


Fig:5.1Performance Metrics Of Adaboost Decision tree Classifier

2.2 Adaboost Random Forest Classifier:

1. The AUC score for AdaBoost RandomForest classifier with 0.948 which is slightly greater than Decision tree classifier
2. Moreover when we look at confusion matrix, False Negative=7 and 'Recall score = 0.897' can be observed which is slightly higher than recall score of decision tree. Thus the performance of the random forest is better than the decision tree AdaBoost.

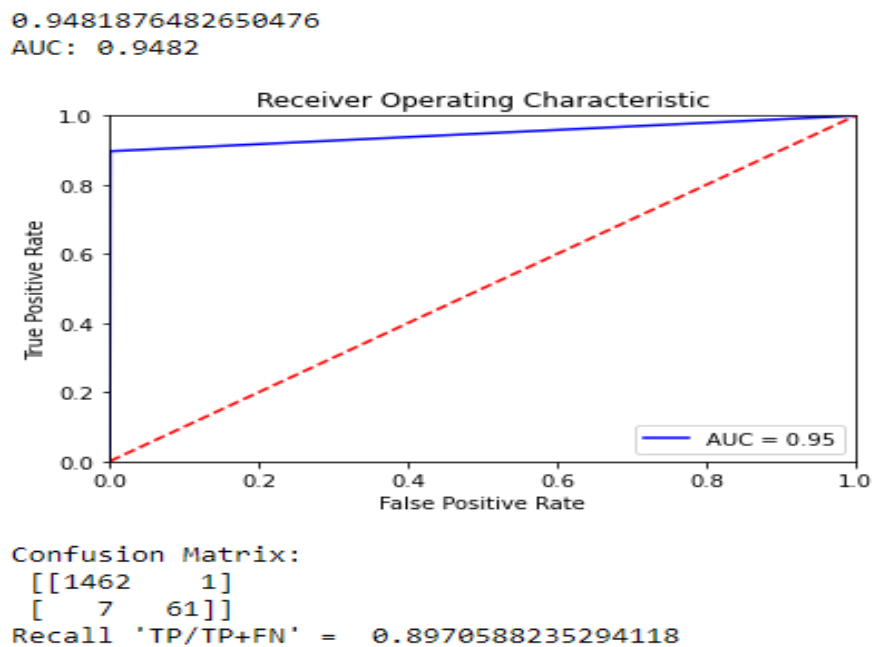


Fig:5.2 Performance Metrics Of Adaboost Random Forest Classifier

2.3 XGBoost model:

1. The XGboost is really good at classifying positive and negative classes where its AUC score is around 0.985 which is slightly greater than Decision tree classifier and also Random Forest classifier.
2. Moreover when we look at confusion matrix, False Negative=6 and 'Recall score = 0.911' can be observed which is slightly higher than recall score of decision tree and random forest classifier.

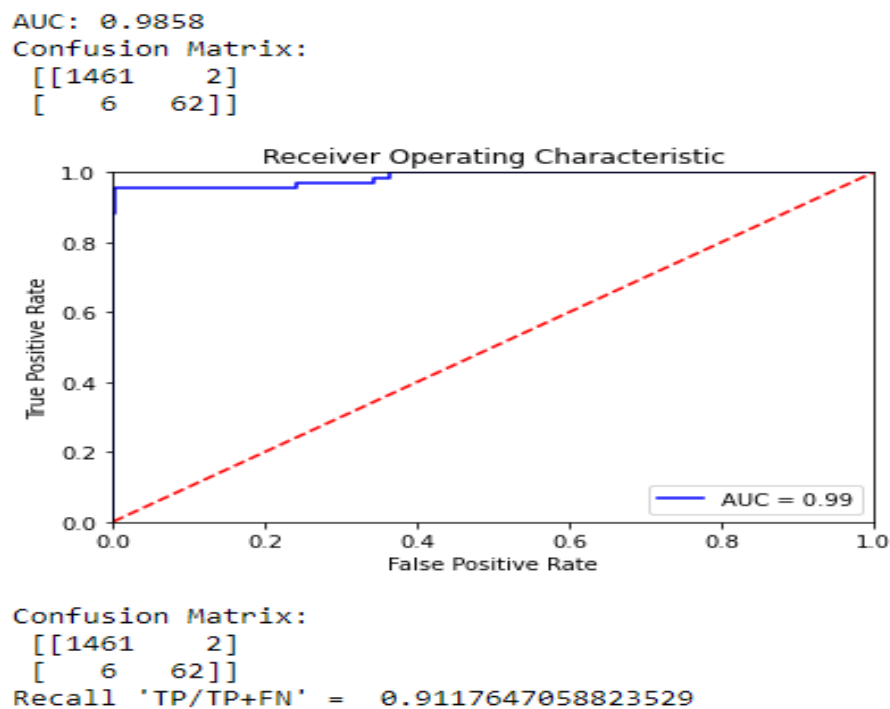


Fig:5.3 Performance Metrics Of XGBoost Classifier

Sl.No	Algorithms	AUC_ROC	Recall
1	Adaboost decision tree classifier	0.94	0.88
2	Adaboost random forest classifier	0.95	0.89
3	XgBoost	0.99	0.91

Fig:5.4:Comparision Of Performance Metrics Of Classifiers

- Since XgBoost model is having higher Recall = '0.91' & auc_score = '0.99' than other algorithms, it can be considered more robust as it has ability to handle class imbalance with Recall score, and deals good with False negative values and penalize it which is important for our task.
- As we definitely would not like the model with higher False Negative values , since its more dangerous to predict as no earthquake while in reality earthquake happened than predicting earthquake will happen given in reality it did not. Since its better safe than sorry! we can allow False positive more than False negative.
- So as XgBoost is having slight less false negatives = '6' than the decision tree and random forest classifier its better suitable for the forecasting.
- Hence we consider XgBoost for forecasting of live data.

CHAPTER 6

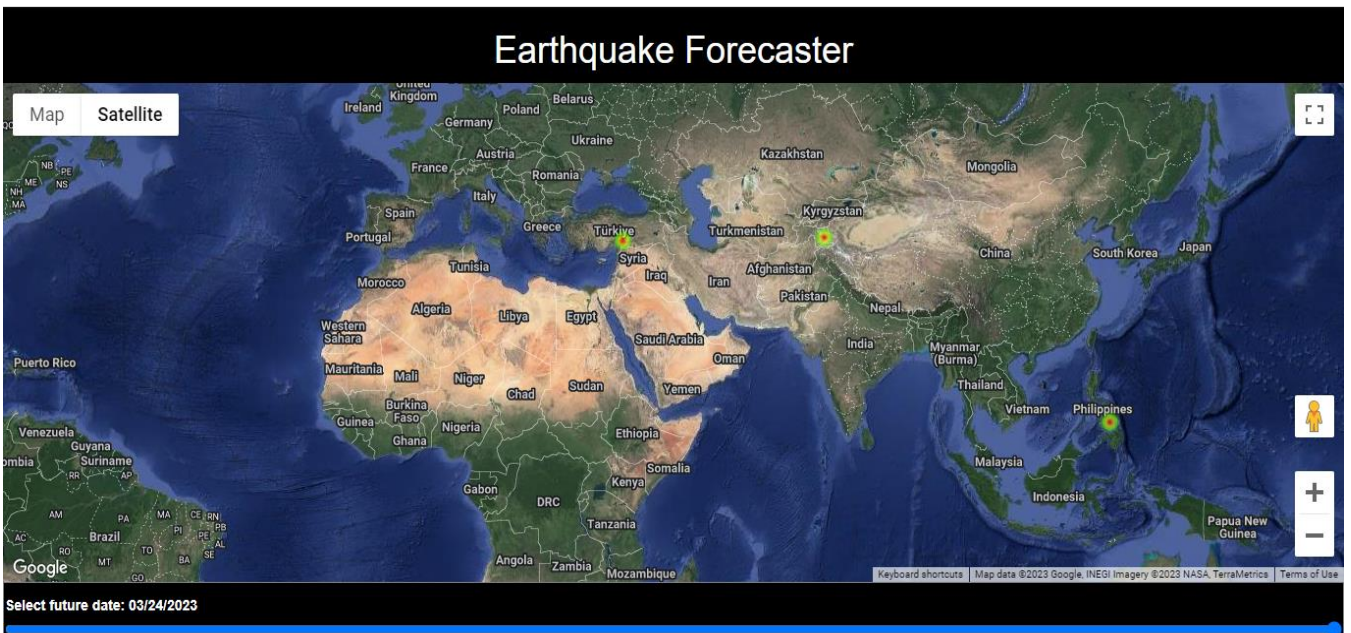
OUTPUTS

OUTPUTS

1. Google map view:



2. Satellite view:



CHAPTER 7

CONCLUSION AND FUTURE SCOPE

1.CONCLUSION

- Earthquake prediction has been marred by repeated failure due to highly uncertain conditions of earth and its surroundings. As we know, earthquakes are one of the worst natural hazards which often turns into disaster causing widespread destruction and loss to human life.
- To overcome this, we are proposing An Automatic Earthquake Forecasting system that is used for forecasting earthquake possible location.
- So after seeing the comparison on auc_roc score, confusion matrix, and recall score of our algorithms, as all the considered algorithms have given similar result with slight differences in their metrics, XgBoost with higher auc_roc score and Recall performs better than other considered boosting algorithms.
- So we consider XgBoost for forecasting the earthquake's possible locations.

2.FUTURE SCOPE

Though XGBoost model has given higher AUC_ROC and better Recall, we believe any work given always has some scope for improvement and in here we could also use RNN or LSTM for time series or rather event series forecasting. LSTMs have hidden memory cells that help in remembering and handling time series or event series data well. Moreover for XgBoost we have just used hyper parameters from already tuned AdaBoost models, but we can also tune XgBoost hyper parameter and find best parameters using GridSearchCV or RandomSearch.

CHAPTER 8

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1.REFERENCES

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